

Committing Security of BBB Secure MACs

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Abstract. Committing security has recently emerged as an essential property for message authentication codes (MACs), driven by applications such as abuse reporting in end-to-end encrypted messaging systems. Traditional notions, such as unforgeability or pseudorandomness, are insufficient in these contexts, prompting the introduction of stronger security notions. Bhaumik et al. (CRYPTO’24) initiated this line of research by formalizing the notions of commitment and context-discovery for MACs and analyzing several standardized birthday-bound secure constructions. We extend this line of work to beyond-birthday-bound (BBB) secure MACs. In particular, we examine the class of constructions identified by Chen et al. (ASIACRYPT’21), which employ two block ciphers and a single block hash function, together with well-studied BBB MACs from the DbHtS paradigm. Our findings depict a heterogeneous picture: while several constructions succumb to simple attacks, others exhibit some resilience, highlighting that committing and context-discovery security for BBB MACs depends strongly on structural design choices.

Keywords: Committing security, Key-robustness, Context-discovery, MAC

1 Introduction

Message Authentication Codes (MACs) are fundamental cryptographic primitives that ensure data integrity and authenticity in symmetric-key settings. They are ubiquitously deployed in secure communication protocols, software update mechanisms, and secure data storage systems. Widely adopted standards such as CMAC [19], and GMAC [20] are valued for their efficiency and strong security under well-established assumptions. However, such schemes generally guarantee security only up to the birthday bound, approximately $2^{n/2}$ queries for a block size of n bits. This bound imposes significant constraints in the domain of lightweight cryptography, where many modern block ciphers are designed for resource-constrained environments such as the Internet of Things (IoT) and employ smaller block sizes, typically 64 or 96 bits, to reduce implementation overhead. Consequently, MAC constructions in these contexts must ensure security beyond the birthday threshold to support practical data limits under a single key.

1.1 Beyond-Birthday-Bound Secure MAC Constructions

Several deterministic MACs that surpass the birthday bound have been proposed. Earlier BBB secure block-cipher-based examples include SUM-ECBC [42], PMAC_Plus [43], 3kf9 [44], GCM-SIV2 [25], and LightMAC_Plus [33], all of which follow the double-block hash-then-sum (DbHtS) paradigm [13]. Where messages are hashed into two blocks, each encrypted, and summed to form the tag. Recently, Chakraborty et al. [8] introduced double-block hash-then-feistel (DbHtF), replacing the sum-of-two n -bit permutations with a 4-round permutation-based Feistel structure [29], achieving optimal n -bit security for 4F-PolyHash-DbHtF, 4F-ECBC-DbHtF, 4F-PHash-DbHtF, and 4F-LightHash-DbHtF.

Another approach based on compression functions for beyond-birthday-bound secure MAC constructions was explored in Multilane HMAC [39], and subsequent works such as [17, 40, 41].

A different approach, introduced by Wegman and Carter [37] based on the work of Gilbert et al. [22], uses a single block hash function and a pseudorandom function (PRF). Following this, Cogliati et al. [12] proposed a beyond-birthday-bound secure MAC against nonce-respecting adversaries, called Encrypted Wegman-Carter with Davies-Meyer (EWCDM), based on a block cipher-based PRF called Encrypted Davies-Meyer (EDM). A similar design was also proposed by Datta et al. [14], called Decrypted Wegman-Carter with Davies-Meyer (DWCDM), which achieves beyond-birthday-bound security against nonce-respecting adversaries. Subsequently, the first beyond-birthday-bound secure MAC against nonce-misuse

(faulty nonce) adversaries was proposed by Dutta et al. [18], called nEHtM, following the design principle of EHtM [32].

Following this design approach, Chen et al. [10] systematically classified MAC constructions using exactly two block cipher calls and a single hash function. Among all variants, only nine achieve beyond-birthday-bound (BBB) security: five based on EDM [12], three on SoP [5], and one on EDMD [31].

Three constructions, $F_{B_1}^{\text{EDM}}$, $F_{B_1}^{\text{SoP}}$, and $F_{B_1}^{\text{EDMD}}$, attain optimal n -bit security with fresh nonces but fail under nonce reuse. The remaining six constructions, $F_{B_2}^{\text{EDM}}$ (a.k.a. EWCDM [12]), $F_{B_3}^{\text{EDM}}$, $F_{B_4}^{\text{EDM}}$, $F_{B_5}^{\text{EDM}}$, $F_{B_2}^{\text{SoP}}$, and $F_{B_3}^{\text{SoP}}$, achieve $3n/4$ -bit security with fresh nonces, of which the last four maintain BBB security even under nonce misuse. A tenth construction, $F_{B_2}^{\text{EDMD}}$, was conjectured to be secure under fresh nonces. Choi et al. [11] later analyzed several constructions in this classification and established stronger security bounds.

1.2 Committing Security

Early notions of *robustness*, requiring that a ciphertext should not verify under multiple keys, were initially formalized for public-key encryption and later adapted to symmetric primitives, laying the conceptual foundation for key- and context-committing Authenticated Encryption with Associated Data (AEAD) [2, 35]. Practical motivations emerged from message franking: Grubbs et al. [23] introduced compactly committing AEAD for verifiable abuse reporting, while Dodis et al. [16] accelerated these constructions via the *encryption* primitive. Subsequent attacks exposed real-world risks of non-committing AEAD, including partitioning-oracle key recovery [28] and tag malleability [3], underscoring the necessity of commitment.

Formally, Bellare and Hoang [4] presented an efficient, provably secure committing AEAD with a clear hierarchy of “commit-to-inputs” goals. Chan and Rogaway [9] proposed a unifying cAE framework, clarifying its relationship to prior robustness notions, while Menda et al. [30] introduced a framework for context-discovery security. The IETF has since documented commitment among desirable AEAD properties, reflecting its practical adoption [7]. Earlier, Farshim et al. [21] established a formal relation between the committing security of AEAD schemes and that of their underlying MACs and encryption components. Independently, Grubbs et al. also highlighted connections with a related property known as receiver binding.

Recent works [15, 27, 36] emphasize that the committing behavior of the authentication component is critical for the overall security of generic AEAD compositions. Beyond AEAD, the widespread independent use of keyed authentication primitives motivates a focused study of their committing security.

In this direction, Bhaumik et al. [6] formalized a suite of security notions capturing the committing behavior of MACs, including key-committing (CMT_k), context-committing (CMT) following [4], context-discovery (CDY) and its key-hiding variant (CDY_k) following [30], and key robustness (RBT_k). These definitions aim to model attacks wherein an adversary reuses or substitutes keys, contexts, or associated data, and still manages to produce valid authentication tags. Analysis under these definitions revealed that several widely used MACs, including HMAC and CMAC, remain vulnerable despite classical UF-CMA security. Bhaumik et al. also proved CMT and CDY security for HMAC with fixed-length keys [24], and proposed a fix for CBC-MAC and CMAC using the Davies-Meyer transformation to achieve CMT_k and CDY security. This motivates a systematic study of the committing security of existing BBB-secure MACs. Following this direction, [26, 38] analyze the committing security of several existing MAC constructions, including PMAC, LightMAC, UMAC, and SMAC.

1.3 Our Contribution

Building on the foundational work of Bhaumik et al. [6], we systematically investigate the committing security of several beyond-birthday-bound (BBB) secure MAC constructions under the notions CMT_k , CMT, CDY_k , CDY, and RBT_k . We first analyze all BBB-secure nonce-based constructions using two block cipher calls and a single hash function, following the classification of Chen et al. [10]. EDM-based constructions are examined in Section 4, EDMD-based constructions in Section 5, and SoP-based constructions in Section 6, with results for nEHtM presented in Section 7. We then extend our analysis in Section 8 to BBB-secure MACs following the double-block hash-then-sum paradigm, including 3kf9, PMAC_Plus, SUM-ECBC, and LightMAC_Plus.

⁴ Success probability of the attack is 2^{-l} , where l is the counter length for each block.

⁵ Success probability of the attack is 2^{-2l} , where l is the counter length for each block.

Table 1: Summary of security results. CMT_k , CMT , RBT_k , CDY_k , and CDY mean key-committing, context-committing, key-robustness, key-discovery, and context-discovery security, respectively. The symbols " $\times$ " and "?" indicate not secure and unknown status, respectively. We use $(\dagger)$ to denote a tight bound, and $(\star)$ for a non-tight bound with an attack.

Construction	CMT	CMT_k	RBT_k	CDY	CDY_k	Reference
$F_{B_1}^{\text{EDM}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 4.1
$F_{B_2}^{\text{EDM}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/3}(\dagger)$	$2^{n/3}(\dagger)$	Sect. 4.2
$F_{B_3}^{\text{EDM}}$	$\times$	$\times$	$2^{n/4}(\star)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 4.3
$F_{B_4}^{\text{EDM}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/3}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 4.4
$F_{B_5}^{\text{EDM}}$	$\times$	$\times$	$2^{n/4}(\star)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 4.5
$F_{B_1}^{\text{EDMD}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 5.1
$F_{B_1}^{\text{SoP}}$	$\times$	$\times$	$\times$	$2^{n/3}(\dagger)$	$2^{n/3}(\dagger)$	Sect. 6.1
$F_{B_2}^{\text{SoP}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/3}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 6.2
$F_{B_3}^{\text{SoP}}$	$\times$	$\times$	$2^{n/4}(\dagger)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 6.3
nEHtM	$2^{n/4}(\star)$	$2^{n/4}(\star)$	$2^{n/4}(\star)$	$2^{n/2}(\dagger)$	$2^{n/2}(\dagger)$	Sect. 7
3kf9	$\times$	$\times$	$\times$	$\times$	?	Sect. 8.1
PMAC_Plus	$\times$	$\times$	$\times$	$\times$	?	Sect. 8.2
SUM-ECBC	$\times$	$\times$	$\times$	?	?	Sect. 8.3
LightMAC_Plus	$\times$	$\times$	$\times^4$	$\times^5$	?	Sect. 8.4
4F-PHash-DbHtF	$\times$	$\times$	?	$\times$	?	Sect. E.2

All our analyses are carried out under the assumption that the underlying block cipher is an ideal cipher, and the committing security is reduced to the collision resistance and preimage resistance security of the hash function. Table 1 consolidates the results, revealing a diverse spectrum of committing security. Among these BBB-secure constructions, committing security varies noticeably across the different notions, reflecting the subtle trade-offs in their design.

Security Model and Assumptions. The MAC security of these constructions is established in [10, 11] under standard assumptions as AXU of the hash function together with the conventional security of the block cipher. Thus, authenticity is justified in the standard model. To obtain committing security, however, additional guarantees are required. In particular, we rely on chosen-key security of the hash function: collision resistance and preimage resistance, and we analyze in the ideal-cipher model.

Throughout the proofs, the adversary is given access to the ideal cipher E , while the hash function H and the ideal cipher E are assumed to be sampled independently and to have no interaction. This independence is crucial for classifying the adversary's oracle queries according to their ordering, which underlies our case analysis. If H and E were not independent, correlations between their queries could invalidate this separation, and a dedicated hybrid analysis would be required to ensure that such dependencies do not compromise security.

Throughout all the proofs, we reduce the advantage of a committing adversary $\mathcal{A}$ to that of a collision-finding adversary $\mathcal{A}_{cr}$ and/or a preimage-finding adversary $\mathcal{A}_{pre}$ against the hash H . In the reductions, we assume that $\mathcal{A}_{cr}$ and $\mathcal{A}_{pre}$ make the same hash queries as the committing adversary $\mathcal{A}$.

2 Preliminaries

Notation. For a set $\mathcal{X}$, $X \leftarrow \$ \mathcal{X}$ denotes that X is sampled uniformly at random from $\mathcal{X}$. We write $X \leftarrow Y$ to denote that Y is assigned to the variable X . We denote an empty set as $\emptyset$. We say two sets $\mathcal{X}$ and $\mathcal{Y}$ are disjoint if $\mathcal{X} \cap \mathcal{Y} = \emptyset$ and we denote their union as $\mathcal{X} \sqcup \mathcal{Y}$ (which we refer to as *disjoint union*).

For any natural number n , $\{0, 1\}^n$ denotes the set of all bit strings of length n and $\{0, 1\}^*$ denotes the set of all binary strings of arbitrary finite length. Sometimes, we call an element of $\{0, 1\}^n$ a *block*. For $X, Y \in \{0, 1\}^n$, we write $X \oplus Y$ to denote the bitwise xor of X and Y . For any element $X \in \{0, 1\}^*$, we write $|X|$ to denote the number of bits of X and for any two $X, Y \in \{0, 1\}^*$, we write $X \| Y$ to denote the concatenation of X followed by Y . We also represent the concatenation of two arbitrary strings X, Y as (X, Y) . For any natural number q , $[q]$ denotes the set $\{1, \dots, q\}$. For integers $1 \leq b \leq a$, $(a)_b$ denotes $a(a-1) \cdots (a-b+1)$, where $(a)_0 = 1$ by convention. For a positive non-zero integer i , $\langle i \rangle_n$ denotes the n -bit binary representation of integer i , and $\text{lsb}_t(S)$ denotes the least significant t bits of a binary string S .

2.1 Message Authentication Code

A message authentication code $\text{MAC} = (\text{MAC.TGen}, \text{MAC.TVer})$ is a pair of algorithms such that:

- The tag generation algorithm $\text{MAC.TGen}: \mathcal{K}_m \times \mathcal{M} \rightarrow \mathcal{T}$ takes as input a secret key $K \in \mathcal{K}_m$ and a message $M \in \mathcal{M}$, and returns a tag $T \in \mathcal{T}$.
- The verification algorithm $\text{MAC.TVer}: \mathcal{K}_m \times \mathcal{M} \times \mathcal{T} \rightarrow \{0, 1\}$ takes as input a secret key $K \in \mathcal{K}_m$, a message $M \in \mathcal{M}$, and a tag $T \in \mathcal{T}$, and returns a bit $b \in \{0, 1\}$, where $b = 0$ indicates an invalid tag.

We refer to the associated sets $\mathcal{K}_m, \mathcal{M}$, and $\mathcal{T}$ as the MAC key space, message space, and tag space, respectively. We require every MAC to satisfy correctness; namely, for all $(K, M) \in (\mathcal{K}_m, \mathcal{M})$, it must hold that $\text{MAC.TVer}(K, M, \text{MAC.TGen}(K, M)) = 1$.

Nonce-Based MAC. Similar to a MAC, a nonce-based message authentication code $\text{MAC} = (\text{MAC.TGen}, \text{MAC.TVer})$ is a pair of deterministic algorithms that take an additional input $N \in \mathcal{N}$ called a nonce. That is, $\text{MAC.TGen}: \mathcal{K}_m \times \mathcal{N} \times \mathcal{M} \rightarrow \mathcal{T}$ and $\text{MAC.TVer}: \mathcal{K}_m \times \mathcal{N} \times \mathcal{M} \times \mathcal{T} \rightarrow \{0, 1\}$. Again, we require the correctness property: $\text{MAC.TVer}(K, N, M, \text{MAC.TGen}(K, N, M)) = 1$, for all $(K, N, M) \in (\mathcal{K}_m, \mathcal{N}, \mathcal{M})$.

2.2 Context-Discovery and Committing Security Notions for MACs

In [6], the authors formalize a suite of commitment and context-discovery security notions for message authentication codes (MACs). In the following security games, let $\text{MAC} = (\text{MAC.TGen}, \text{MAC.TVer})$ denote a nonce-based message authentication code. Define a randomized algorithm $\mathcal{S}$, referred to as the *context-selector*, which takes no input and outputs a challenge value that is provided to the adversary at the start of the game. The security notions are defined via the security games shown in Figure 1. We write $\text{Adv}_{\text{MAC}}^{\text{Game } G}(\mathcal{A})$ to denote the probability that $\mathcal{A}$ wins the corresponding game G .

Game CMT	Game CMT _k	
$(K_1, N_1, M_1), (K_2, N_2, M_2) \leftarrow \mathcal{A}()$	$(K_1, N_1, M_1), (K_2, N_2, M_2) \leftarrow \mathcal{A}()$	
if $(K_1, N_1, M_1) = (K_2, N_2, M_2)$	if $(K_1, N_1, M_1) = (K_2, N_2, M_2)$	
return 0	return 0	
$T_1 \leftarrow \text{MAC.TGen}(K_1, N_1, M_1)$	$T_1 \leftarrow \text{MAC.TGen}(K_1, N_1, M_1)$	
$T_2 \leftarrow \text{MAC.TGen}(K_2, N_2, M_2)$	$T_2 \leftarrow \text{MAC.TGen}(K_2, N_2, M_2)$	
return $(T_1 = T_2)?1 : 0$	return $(K_1 \neq K_2 \wedge T_1 = T_2)?1 : 0$	
Game RBT _k	Game CDY	Game CDY _k
$(K_1, K_2, N, M) \leftarrow \mathcal{A}()$	$T \leftarrow \mathcal{S}()$	$(N, M, T) \leftarrow \mathcal{S}()$
$T_1 \leftarrow \text{MAC.TGen}(K_1, N, M)$	$(K, N, M) \leftarrow \mathcal{A}(T)$	$K \leftarrow \mathcal{A}(N, M, T)$
$T_2 \leftarrow \text{MAC.TGen}(K_2, N, M)$	$T' \leftarrow \text{MAC.TGen}(K, N, M)$	$T' \leftarrow \text{MAC.TGen}(K, N, M)$
return $(K_1 \neq K_2 \wedge T_1 = T_2)?1 : 0$	return $(T' = T)?1 : 0$	return $(T' = T)?1 : 0$

Fig. 1: Security games CMT, CMT_k, RBT_k, CDY and CDY_k.

Relation between commitment and context-discovery notions for MACs. In [6], the authors establish the following relations between previously identified security notions for a MAC scheme: committing security and context-discovery security.

Lemma 1. *Given any MAC scheme, the following security relations hold:*

1. *if it is CMT secure then it is CMT_k secure,*
2. *if it is CMT_k secure then it is RBT_k secure,*
3. *if it is CDY secure then it is CDY_k secure for a family of context-selectors,*
4. *if it is CMT secure then it is CDY_k secure for a family of context-selectors.*

2.3 Security Notions of a Hash Function

Let $H : \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function. We define the security notions of collision resistance and preimage resistance via the following games.

Collision resistance Game	Preimage resistance Game
$(X, X') \leftarrow \mathcal{A}^H()$	$(X, S) \leftarrow \mathcal{A}^H()$
return $(X \neq X' \wedge H(X) = H(X')) ? 1 : 0$	if $\mathcal{A}$ has queried X to H
	return 0
	return $(S \leq s \wedge H(X) \in S) ? 1 : 0$

Fig. 2: Collision resistance and Preimage resistance Games of a hash function.

For an adversary $\mathcal{A}$, the collision resistance advantage and the preimage resistance advantage of a hash function H are defined as follows.

$$\begin{aligned} \text{Adv}_H^{\text{CR}}(\mathcal{A}) &:= \Pr [(X, X') \leftarrow \mathcal{A}^H(\cdot) : X \neq X', H(X) = H(X')] , \\ \text{Adv}_H^{\text{Pre}}(\mathcal{A}; s) &:= \Pr [(X, S) \leftarrow \mathcal{A}^H(\cdot) : H(X) \in S \wedge |S| \leq s] . \end{aligned}$$

3 Results on DM and EDM Constructions

In this section, we recall the Davies-Meyer (DM) and Encrypted Davies-Meyer (EDM) constructions, together with their bounds on preimage resistance and collision resistance. These bounds will be used in the subsequent analysis.

3.1 Preimage and Collision Resistance Bounds for DM

Let $E : \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be an ideal block cipher. The Davies-Meyer construction $\text{DM} : \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ is defined as

$$\text{DM}[E](K, U) = E_K(U) \oplus U.$$

In [6, Lemma 1], the authors established the following bounds on the collision resistance and preimage resistance of the DM construction.

Lemma 2. *For any adversary making at most $q_p \leq 2^{n-1}$ ideal-cipher queries to E and E^{-1} , we have*

$$\text{Adv}_{\text{DM}}^{\text{coll}}(\mathcal{A}) \leq \frac{q_p^2}{2^n}, \quad \text{Adv}_{\text{DM}}^{\text{epre}}(\mathcal{A}) \leq \frac{2q_p}{2^n}.$$

3.2 Preimage and Collision Resistance Bounds for EDM

Let $E : \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be an ideal block cipher. The Encrypted Davies-Meyer construction is defined as

$$\text{EDM}[E](K_1, K_2, N) = E_{K_2}(E_{K_1}(N) \oplus N).$$

Lemma 3 (Fixed-output preimage and collision for 0^n). *There exist deterministic polynomial-time oracle algorithms that*

1. *output (K_1, K_2, N) with $\text{EDM}(K_1, K_2, N) = 0^n$ using a single inverse-oracle query; and*
2. *output (K_1, K_2, N) and (K'_1, K'_2, N') such that $\text{EDM}(K_1, K_2, N) = \text{EDM}(K'_1, K'_2, N') = 0^n$ with $(K_1, K_2) \neq (K'_1, K'_2)$, using two inverse-oracle queries.*

Proof. For (1): fix any key $K \in \{0, 1\}^\kappa$ and set $N = E_K^{-1}(0^n)$. Then

$$\text{EDM}(K, K, N) = E_K(E_K(N) \oplus N) = E_K(N) = 0^n.$$

For (2): pick distinct keys $K \neq K'$, set $N = E_K^{-1}(0^n)$ and $N' = E_{K'}^{-1}(0^n)$. As above,

$$\text{EDM}(K, K, N) = 0^n = \text{EDM}(K', K', N'),$$

and $(K, K, N) \neq (K', K', N')$, yielding a collision at output 0^n .

Lemma 4. (Collision at 0^n with same N and different keys.) *Let an adversary make at most q_p queries to the ideal block-cipher oracle, with $q_p \leq 2^{n-1}$. Then the probability that the adversary outputs two distinct key-tuples $(K_1, K_2) \neq (K'_1, K'_2)$ with the same input N such that*

$$\text{EDM}(K_1, K_2, N) = \text{EDM}(K'_1, K'_2, N) = 0^n$$

is at most $4q_p^2/2^n$.

Proof. Fix two candidate inputs (K_1, K_2, N) and (K'_1, K'_2, N) with $(K_1, K_2) \neq (K'_1, K'_2)$, and let $V = E_{K_1}(N) \oplus N$ and $V' = E_{K'_1}(N) \oplus N$.

Case 1: $(K_2, V) = (K_1, N)$ and $(K'_2, V') = (K'_1, N)$. Then

$$\text{EDM}(K_1, K_1, N) = E_{K_1}(N), \quad \text{EDM}(K'_1, K'_1, N) = E_{K'_1}(N).$$

Thus both equal 0^n only if $E_{K_1}(N) = E_{K'_1}(N) = 0^n$. In the ideal-cipher model, for distinct $K_1 \neq K'_1$ and fixed N , the values $E_{K_1}(N)$ and $E_{K'_1}(N)$ are independent and uniform in $\{0, 1\}^n$. Hence, the probability of this event across at most q_p^2 pairs of queries is bounded by $2q_p^2/2^n$.

Case 2: Otherwise, assume $(K_2, V) \neq (K_1, N)$. Then

$$\text{EDM}(K_1, K_2, N) = 0^n \iff E_{K_1}(N) \oplus N = E_{K_2}^{-1}(0^n).$$

For fixed (K_1, K_2, N) the left-hand side is uniform over $\{0, 1\}^n$ and independent of the right-hand side, so the equality holds with probability $1/2^n$. Across at most q_p^2 pairs, this contributes at most $2q_p^2/2^n$.

Combining with Case 1, the total success probability is at most $4q_p^2/2^n$.

Lemma 5. (Collision resistance for $W \neq 0^n$.) *Let an adversary make at most q_p queries to the ideal block-cipher oracle, with $q_p \leq 2^{n-1}$. Then the probability that the adversary outputs two distinct tuples $(K_1, K_2, N) \neq (K'_1, K'_2, N')$ such that*

$$\text{EDM}(K_1, K_2, N) = \text{EDM}(K'_1, K'_2, N') = W \neq 0^n$$

is at most $6q_p^2/2^n$.

Proof. The condition

$$E_{K_2}(E_{K_1}(N) \oplus N) = E_{K'_2}(E_{K'_1}(N') \oplus N') = W$$

leads to two cases.

Case 1: $K_2 = K'_2$. Then $E_{K_1}(N) \oplus N = E_{K'_1}(N') \oplus N'$ for distinct $(K_1, N) \neq (K'_1, N')$. In the ideal-cipher model these outputs are independent uniforms, so the probability is at most $2q_p^2/2^n$.

Case 2: $K_2 \neq K'_2$. Assume the query on (K_2, V) precedes that on (K'_2, V') . A collision $E_{K_2}(V) = E_{K'_2}(V') = W$ can occur in two ways: (i) *Forward:* the adversary queries $E_{K'_2}(V')$ and gets W , with probability at most $2q_p^2/2^n$. (ii) *Inverse:* the adversary queries $E_{K_2}^{-1}(W) = V'$, requiring $E_{K'_1}(N') \oplus N' = V'$, which again holds with probability at most $2q_p^2/2^n$.

Adding both cases, the overall success probability is bounded by $6q_p^2/2^n$.

Lemma 6 (Preimage finding for $W \neq 0^n$). *Let $W \neq 0^n$ be any fixed value. An adversary making at most q_p queries to the ideal block-cipher oracle, with $q_p \leq 2^{n-1}$, has probability at most $2q_p^2/2^n$ of finding a tuple (K_1, K_2, N) such that $\text{EDM}(K_1, K_2, N) = W$.*

Proof. If $(K_1, N) = (K_2, V)$ with $V = E_{K_1}(N) \oplus N$, then $\text{EDM}(K_1, K_2, N) = 0^n$, contradicting $W \neq 0^n$. Hence necessarily $(K_1, N) \neq (K_2, V)$. In this case,

$$\text{EDM}(K_1, K_2, N) = W \iff E_{K_1}(N) \oplus N = E_{K_2}^{-1}(W).$$

For fixed (K_1, K_2, N) , the left-hand side is uniform in $\{0, 1\}^n$ and independent of the right-hand side, so the equality holds with probability $1/2^n$. With at most q_p^2 possible query pairs and considering both forward and inverse directions, the overall success probability is bounded by $2q_p^2/2^n$.

4 Analysis of Nonce-based MACs (based on EDM)

Chen et al. [10] introduced nonce-based pseudorandom functions (PRFs) based on the Encrypted Davies-Meyer (EDM) construction, denoted as F^{EDM} . Let $\kappa, \kappa_h, n \in \mathbb{N}$, $E: \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher and $H: \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function. They proposed a generic construction

$$F_{B_x}^{\text{EDM}}: \{0, 1\}^{2\kappa} \times \{0, 1\}^{\kappa_h} \times \{0, 1\}^n \times \{0, 1\}^* \rightarrow \{0, 1\}^n,$$

defined as

$$\begin{aligned} F_{B_x}^{\text{EDM}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) \\ = E_{K_2}(E_{K_1}(N \oplus b_1 \cdot H(K_h, M)) \oplus N \oplus b_2 \cdot H(K_h, M)) \oplus b_3 \cdot H(K_h, M), \end{aligned}$$

where $B_x \in \{B_0, \dots, B_7\}$ is the 3-bit representation of x , denoted as $(b_1, b_2, b_3) \in \{0, 1\}^3$. The authors showed that among the eight possible constructions, five-namely, $F_{B_1}^{\text{EDM}}$, $F_{B_2}^{\text{EDM}}$, $F_{B_3}^{\text{EDM}}$, $F_{B_4}^{\text{EDM}}$, and $F_{B_5}^{\text{EDM}}$ -achieve beyond-birthday-bound (BBB) MAC security against nonce-respecting adversaries.

4.1 $F_{B_1}^{\text{EDM}}$

The function $F_{B_1}^{\text{EDM}}$ is a Wegman-Carter construction that uses EDM as its underlying PRF. It provides optimal n -bit MAC security against nonce-respecting adversaries but becomes completely insecure if the nonce is reused. Let E be a block cipher and H be a hash function, $F_{B_1}^{\text{EDM}}$ is defined as

$$F_{B_1}^{\text{EDM}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N) \oplus N) \oplus H(K_h, M).$$

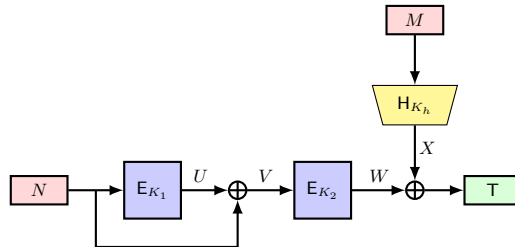


Fig. 3: $F_{B_1}^{\text{EDM}}$

4.1.1 CMT and CMT_k Security of $F_{B_1}^{\text{EDM}}$ Here, we present a key-committing attack against $F_{B_1}^{\text{EDM}}$ construction. The attack works as follows: fix any (K_h, M) and then exploit Lemma 3, which guarantees two distinct triples $(K, K, N), (K', K', N')$, where $K \neq K'$ with $\text{EDM}(K, K, N) = \text{EDM}(K', K', N') = 0^n$. Xoring with $H(K_h, M)$ preserves this collision, so the inputs (K, K, K_h, N, M) and (K', K', K_h, N', M) yield the same tag. The formal attack description is given in Figure 4.

- 1 : Select any $K \in \{0, 1\}^\kappa$ and compute $N = E_K^{-1}(0^n)$;
- 2 : Select another $K' \in \{0, 1\}^\kappa$ such that $K' \neq K$ and compute $N' = E_{K'}^{-1}(0^n)$;
- 3 : Select any $K_h \in \{0, 1\}^{\kappa_h}$ and $M \in \{0, 1\}^*$;
- 4 : Output the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) ;

Fig. 4: CMT_k adversary against $F_{B_1}^{\text{EDM}}$

Therefore, the construction $F_{B_1}^{\text{EDM}}$ is not CMT_k secure, and using Lemma 1, we can conclude that it is also not CMT secure.

4.1.2 RBT_k security of $F_{B_1}^{\text{EDM}}$

Theorem 1. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key robustness advantage satisfies:

$$\text{Adv}_{F_{B_1}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p q_h) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h) + \frac{6q_p^2}{2^n} + \frac{2q_h^2 q_p}{2^n} + \frac{2q_p^2 q_h^2}{2^n}.$$

Proof. Let $\mathcal{A}$ output $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that:

$$(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h).$$

Now, let $T = E_{K_2}(E_{K_1}(N) \oplus N) \oplus H(K_h, M)$ and $T' = E_{K'_2}(E_{K'_1}(N) \oplus N) \oplus H(K'_h, M)$. Therefore, the adversary wins iff it can satisfy $T = T'$, i.e.

$$F_{B_1}^{\text{EDM}}(K_1, K_2, K_h, N, M) = F_{B_1}^{\text{EDM}}(K'_1, K'_2, K'_h, N, M).$$

We analyze the adversary's success probability across two cases.

Case 1: $K_h = K'_h$: In this case, $H(K_h, M) = H(K'_h, M)$, so the equality $T = T'$ implies:

$$\text{EDM}(K_1, K_2, N) = \text{EDM}(K'_1, K'_2, N).$$

Since $K_h = K'_h$, so it must be $(K_1, K_2) \neq (K'_1, K'_2)$, and the above equality boils down to a collision of $\text{EDM}(K_1, K_2, N)$, which occurs with probability at most $2q_p^2/2^n$ following Lemma 4.

Case 2: $K_h \neq K'_h$: We further divide this case into two subcases:

- (a) Let the adversary finds $K_h \neq K'_h$ such that $H(K_h, M) = H(K'_h, M)$ occurs. Then, the adversary's success probability is upper-bounded by the hash collision advantage $\text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$. Where $\mathcal{A}_{cr}$ is a collision finding adversary for H that uses the same hash query as $\mathcal{A}$.
- (b) Let $H(K_h, M) \neq H(K'_h, M)$. Let

$$V := E_{K_1}(N) \oplus N, \quad W := E_{K_2}(V), \quad V' := E_{K'_1}(N) \oplus N, \quad W' := E_{K'_2}(V').$$

Then the equality $T = T'$, i.e., $W \oplus H(K_h, M) = W' \oplus H(K'_h, M)$ implies $W \neq W'$ and hence $(K_1, K_2) \neq (K'_1, K'_2)$. Now, we further analyze this subcase depending on the structure of primitive queries:

- (i) $(K'_1, N) = (K'_2, V')$: Then $W' = 0^n$, so $W = H(K_h, M) \oplus H(K'_h, M)$.
 - If the final query among $H(K_h, M)$, $H(K'_h, M)$, and $W (= E_{K_2}(V))$ is a forward ideal cipher query for W , then the event occurs with probability at most $2q_h^2 q_p / 2^n$.
 - If the final query is an inverse ideal-cipher query, then the adversary must satisfy $E_{K_1}(N) \oplus N = E_{K_2}^{-1}(W)$ with $W \neq 0^n$, which occurs with probability at most $2q_p^2 / 2^n$.
 - Otherwise, without loss of generality let $H(K_h, M)$ be the final query. The adversary succeeds only if $H(K_h, M) \in \{W \oplus H(K_h, M)\}$; since there are at most q_p choices for W and q_h hash outputs, this target set has size at most $q_p q_h$. Hence, the success probability is upper-bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p q_h)$, where $\mathcal{A}_{pre}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

- (ii) $(K'_1, N) \neq (K'_2, V')$: Then $W \oplus W' = H(K_h, M) \oplus H(K'_h, M)$.
- If the final query among $W (= E_{K_2}(V))$, $W' (= E_{K'_2}(V'))$, $H(K_h, M)$, and $H(K'_h, M')$ is a forward ideal-cipher query, then the event occurs with probability at most $2q_h^2 q_p^2 / 2^n$.
 - If the final query is an inverse ideal-cipher query (w.l.o.g. involving (K'_2, V', W')), then it requires $E_{K'_1}(N) \oplus N = E_{K'_2}^{-1}(W')$, which occurs with probability at most $2q_p^2 / 2^n$.
 - Otherwise, without loss of generality, suppose that $H(K_h, M)$ is the final query. The adversary succeeds only if $H(K_h, M) \in \{W \oplus W' \oplus H(K'_h, M')\}$. Since there are at most q_p possible values for each of W and W' and at most q_h possible hash outputs, the corresponding target set $\{W \oplus W' \oplus H(K'_h, M')\}$ has size at most $q_p^2 q_h$. Hence, the adversary's success probability is upper-bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h)$, where $\mathcal{A}'_{pre}$ is a preimage-finding adversary issuing the same hash queries as $\mathcal{A}$.

Combining all the above cases, the theorem follows.

Matching attack for RBT_k on $F_{B_1}^{\text{EDM}}$: We now demonstrate the tightness of the RBT_k bound for $F_{B_1}^{\text{EDM}}$ as established in the preceding theorem. The attack proceeds as follows:

- Fix a key K_1 and a nonce N . Then compute $V = N \oplus E_{K_1}(N)$.
- Select $2^{n/4}$ distinct values for K_2^i . For each $i \in [1, 2^{n/4}]$, compute $W_i = E_{K_2^i}(V)$.
- Independently, select $2^{n/4}$ distinct values for K_h , and for each $i \in [2^{n/4}]$, compute $X_i = H(K_h^i, M)$.
- Find indices $a, b, c, d \in [2^{n/4}]$ such that $W_a \oplus X_b = W_c \oplus X_d$.
- Output the tuple $((K_1, K_2^a, K_h^b), (K_1, K_2^c, K_h^d), N, M)$.

Since each W_i and X_i is sampled uniformly at random and independently, the success probability of the attack is at least $1 - e^{-1}$.

4.1.3 CDY and CDY_k security of $F_{B_1}^{\text{EDM}}$

Theorem 2. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context selector S does not access $F_{B_1}^{\text{EDM}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies

$$\text{Adv}_{F_{B_1}^{\text{EDM}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p) + \frac{2q_h q_p}{2^n} + \frac{2q_p^2}{2^n}.$$

Proof. Let T denote the challenge tag. An adversary outputs (K_1, K_2, K_h, N, M) and succeeds if

$$F_{B_1}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T,$$

that is, if $V = E_{K_1}(N)$, $W = E_{K_2}(V)$, $X = H(K_h, M)$, and $T = W \oplus X$.

Case 1: $(K_1, N) = (K_2, V)$. Then $W = 0^n$ and $T = X = H(K_h, M)$. Hence, the success probability of $\mathcal{A}$ is upper-bounded by the preimage-finding advantage $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; 1)$, where $\mathcal{A}_{pre}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

Case 2: $(K_1, N) \neq (K_2, V)$. In this case, $T = E_{K_2}(V) \oplus H(K_h, M)$.

- 2a: If (K_2, V, W) is queried after $H(K_h, M)$ and is a forward ideal-cipher query, then the probability of matching both components over at most q_p ideal-cipher queries and q_h hash queries is at most $2q_p q_h / 2^n$.
- 2b: If (K_2, V, W) is queried after $H(K_h, M)$ and is an inverse ideal-cipher query, then $E_{K_1}(N) \oplus N = V = E_{K_2}^{-1}(W)$, which occurs with probability at most $2q_p^2 / 2^n$.
- 2c: Otherwise, $H(K_h, M)$ is queried after (K_2, V, W) . The adversary succeeds only if $H(K_h, M) \in \{W \oplus T\}$. Since there are at most q_p possible values of W , the success probability is upper-bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p)$, where $\mathcal{A}'_{pre}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

Summing over all cases yields the claimed bound.

Therefore, $F_{B_1}^{\text{EDM}}$ achieves CDY security against $O(2^{n/2})$ queries. By Lemma 1, it further satisfies CDY_k security for the same query bound.

Matching CDY_k Attack on $F_{B_1}^{\text{EDM}}$: Here, we present a matching attack on CDY_k , which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_1}^{\text{EDM}}$. Let $(N, M, \mathbf{T})$ denote the challenge tuple. The adversary proceeds as follows:

- Select $K_h \in \{0, 1\}^{\kappa_h}$ and compute $X = \text{H}(K_h, M)$.
- Select $2^{n/2}$ many distinct values for K_1^i and for each $i \in [2^{n/2}]$, compute $U_i = \text{E}_{K_1^i}(N)$.
- Select $2^{n/2}$ many distinct values for K_2^j and for each $j \in [2^{n/2}]$, compute $V_j = \text{E}_{K_2^j}^{-1}(X \oplus \mathbf{T})$.
- If $U_i \oplus V_j = N$ for some i, j , output (K_1^i, K_2^j, K_h) .

Since U_i and V_j are independent and uniformly random in $\{0, 1\}^n$, the probability that a pair satisfies $U_i \oplus V_j = N$ is at least $1 - e^{-1}$ by the birthday bound.

4.2 $F_{B_2}^{\text{EDM}}$

The function $F_{B_2}^{\text{EDM}}$ is the EWCDM construction of Cogliati and Seurin [12], which is shown to achieve $2n/3$ -bit MAC security against nonce-respecting adversaries. Recently, Choi et al. [11] refined this result (assuming that the underlying hash function is xor-collision resistant), establishing an optimal security bound and proving its tightness. Let E be a block cipher and H be a hash function, $F_{B_2}^{\text{EDM}}$ is defined as

$$F_{B_2}^{\text{EDM}}[\text{E}_{K_1}, \text{E}_{K_2}, \text{H}_{K_h}](N, M) = \text{E}_{K_2}(\text{E}_{K_1}(N) \oplus N \oplus \text{H}(K_h, M)).$$

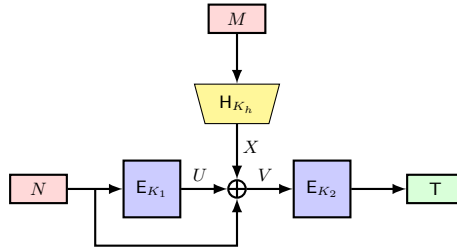


Fig. 5: $F_{B_2}^{\text{EDM}}$

4.2.1 CMT and CMT_k Security of $F_{B_2}^{\text{EDM}}$ Here, we present a key-committing attack against $F_{B_2}^{\text{EDM}}$ construction. The adversary begins by selecting a hash key K_h and a message M , then computes $X = \text{H}(K_h, M)$. Next, the adversary chooses two distinct keys, K and K' , and derives the corresponding nonce values N and N' as $\text{E}_K^{-1}(X)$ and $\text{E}_{K'}^{-1}(X)$, respectively. Finally, the adversary outputs the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) . The complete attack procedure is detailed in Figure 6 and the analysis is described in Appendix A.1.

- 1 : Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$ and compute $X = \text{H}(K_h, M)$;
- 2 : Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
- 3 : Compute $N = \text{E}_K^{-1}(X)$ and $N' = \text{E}_{K'}^{-1}(X)$;
- 4 : Output the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) ;

Fig. 6: CMT_k adversary against $F_{B_2}^{\text{EDM}}$

Therefore, the construction $F_{B_2}^{\text{EDM}}$ is not CMT_k secure, and by Lemma 1, we can conclude that it is also not CMT secure.

4.2.2 RBT_k security of $F_{B_2}^{\text{EDM}}$

Theorem 3. *Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. For any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key-robustness advantage satisfies*

$$\mathbf{Adv}_{F_{B_2}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}) \leq \mathbf{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2) + \mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h) + \frac{4q_p^2}{2^n} + \frac{2q_p^2 q_h}{2^n} + \frac{2q_p^2 q_h^2}{2^n}.$$

Proof. Let $\mathcal{A}$ output $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that

1. $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$, and
2. $F_{B_2}^{\text{EDM}}(K_1, K_2, K_h, N, M) = F_{B_2}^{\text{EDM}}(K'_1, K'_2, K'_h, N, M)$.

Define

$$T := E_{K_2}(E_{K_1}(N) \oplus N \oplus H(K_h, M)) \quad T' := E_{K'_2}(E_{K'_1}(N) \oplus N \oplus H(K'_h, M)).$$

The adversary succeeds if and only if $T = T'$. We distinguish two main cases.

Case 1: $K_2 \neq K'_2$. The ideal-cipher queries involving (K_2, V, T) and (K'_2, V', T') are distinct. Assume that the query involving (K'_2, V', T') occurs after the one involving (K_2, V, T) .

1a: If the query involving (K'_2, V', T') is forward, then the success probability is upper-bounded by $2q_p^2/2^n$.

1b: If the query involving (K'_2, V', T') is inverse, then success implies

$$E_{K'_1}(N) \oplus N \oplus H(K'_h, M) = E_{K'_2}^{-1}(T').$$

We further distinguish two subcases.

1b1: If $H(K'_h, M)$ is the final query among the three relevant queries, then success requires

$$H(K'_h, M) \in \{E_{K'_1}(N) \oplus N \oplus E_{K'_2}^{-1}(T')\}.$$

Since there are at most q_p ideal-cipher queries, the target set has size at most q_p^2 . Hence, the success probability is upper-bounded by $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2)$, where $\mathcal{A}_{pre}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

1b2: Otherwise, one of the remaining ideal-cipher queries is the final query. By the randomness of the ideal cipher, the success probability is upper-bounded by $2q_p^2 q_h / 2^n$.

Case 2: $K_2 = K'_2$. In this case, success implies

$$E_{K_1}(N) \oplus N \oplus H(K_h, M) = E_{K'_1}(N) \oplus N \oplus H(K'_h, M),$$

where $(K_1, K_h) \neq (K'_1, K'_h)$.

2a: $(K_h, M) = (K'_h, M)$. Then success requires a collision in the Davies–Mayer construction,

$$E_{K_1}(N) \oplus N = E_{K'_1}(N) \oplus N.$$

By Lemma 2, the success probability is upper-bounded by $2q_p^2/2^n$.

2b: $(K_h, M) \neq (K'_h, M')$ and $H(K_h, M) = H(K'_h, M')$. In this case, $\mathcal{A}$ yields a collision for H , and the success probability is upper-bounded by $\mathbf{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$, where $\mathcal{A}_{cr}$ makes the same hash queries as $\mathcal{A}$.

2c: $H(K_h, M) \neq H(K'_h, M')$.

2c1: If $H(K'_h, M)$ is the final query among the four relevant queries, then success requires

$$H(K'_h, M) \in \{E_{K_1}(N) \oplus N \oplus H(K_h, M) \oplus E_{K'_1}(N) \oplus N\}.$$

Since there are at most q_p ideal-cipher queries and q_h hash queries, the target set has size at most $q_p^2 q_h$. Hence the success probability is upper-bounded by $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h)$, where $\mathcal{A}'_{pre}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

2c2: Otherwise, let $E_{K'_1}(N)$ be the final query among the relevant queries. By the randomness of the ideal cipher, the success probability is upper-bounded by $2q_p^2 q_h^2 / 2^n$.

Combining all cases yields the stated bound.

We have also demonstrated a matching attack for RBT_k on $F_{B_2}^{\text{EDM}}$ in Appendix A.2.

4.2.3 CDY and CDY_k security of $F_{B_2}^{\text{EDM}}$

Theorem 4. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access any of $F_{B_2}^{\text{EDM}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:

$$\text{Adv}_{F_{B_2}^{\text{EDM}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2) + \frac{2q_p}{2^n} + \frac{2q_p^2 q_h}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h, N, M) such that $F_{B_2}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T$.

The adversary is allowed to make queries to the ideal cipher (both forward and inverse) as well as to the hash oracle. We analyze the success probability of $\mathcal{A}$ by considering the following cases:

Case 1: Let the ideal cipher queries involving (K_1, N, U) and (K_2, V, T) are exactly same. Then, we have

$$N = V \implies N = N \oplus U \oplus H(K_h, M) \implies H(K_h, M) = U = T.$$

So, in this case, the adversary succeeds implies that the equation $H(K_h, M) = T$ is satisfied. Therefore, the adversary's success probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$, where $\mathcal{A}_{\text{pre}}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.

Case 2: The adversary makes forward ideal cipher queries involving K_2 and attempts to find a collision between one of the outputs and the challenge tag T . Using the randomness of ideal cipher E , we have the probability of a successful collision is at most $2q_p/2^n$.

Case 3: The adversary makes inverse ideal cipher queries involving T . In this case, $\mathcal{A}$ succeeds if and only if

$$E_{K_1}(N) \oplus N \oplus H(K_h, M) = E_{K_2}^{-1}(T).$$

Now we will analyze the probability of this event in the following subcases:

- 3a. $H(K_h, M)$ is the final query among three relevant queries. Then the adversary will be successful if $H(K_h, M) \in \{E_{K_1}(N) \oplus N \oplus E_{K_2}^{-1}(T)\}$. As there are at most q_p ideal cipher queries, the cardinality of the target set is at most q_p^2 . Hence, in this case, $\mathcal{A}$'s success probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2)$, where $\mathcal{A}'_{\text{pre}}$ is the preimage finding adversary against H making the same hash query as $\mathcal{A}$.
- 3b. One of the ideal cipher queries among $E_{K_1}(N)$, $E_{K_2}^{-1}(T)$ is the final query. Then, $\mathcal{A}$'s success probability is upper bounded by $2q_p^2 q_h / 2^n$.

Therefore, combining all cases, we have the theorem.

A matching CDY_k attack on $F_{B_2}^{\text{EDM}}$ is described in Appendix A.3. The attack establishes that the above bound is tight for both CDY_k and CDY security of $F_{B_2}^{\text{EDM}}$.

4.3 $F_{B_3}^{\text{EDM}}$

Let E be a block cipher and H be a hash function. The function $F_{B_3}^{\text{EDM}}$ is defined as:

$$F_{B_3}^{\text{EDM}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N) \oplus N \oplus H(K_h, M)) \oplus H(K_h, M).$$

Chen et al. [10] prove that this construction achieves $3n/4$ -bit MAC security against nonce-respecting adversaries. Recently, Choi et al. [11] refined this result (assuming that the underlying hash function is xor-collision resistant), establishing an optimal security bound and proving its tightness.

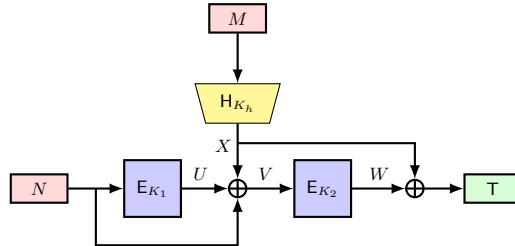


Fig. 7: $F_{B_3}^{\text{EDM}}$

4.3.1 CMT and CMT_k Security of $F_{B_3}^{\text{EDM}}$ We now describe a key-committing attack against the $F_{B_3}^{\text{EDM}}$ construction. The attack proceeds in the same fashion as for $F_{B_2}^{\text{EDM}}$, but we include the full details for completeness. The adversary's strategy is provided in Figure 8 and the analysis is described in Appendix A.4.

1 : Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$ and compute $X = H(K_h, M)$;
 2 : Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
 3 : Compute $N = E_K^{-1}(X)$, $N' = E_{K'}^{-1}(X)$;
 4 : Output the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) ;

Fig. 8: CMT_k adversary against $F_{B_3}^{\text{EDM}}$

Therefore, the construction $F_{B_3}^{\text{EDM}}$ is not CMT_k secure, and by Lemma 1, we can conclude that it is also not CMT secure.

4.3.2 RBT_k security of $F_{B_3}^{\text{EDM}}$

Theorem 5. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. For any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key-robustness advantage satisfies

$$\text{Adv}_{F_{B_3}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h) + \frac{4q_p^2}{2^n} + \frac{2q_p^2 q_h}{2^n} + \frac{2q_p^2 q_h^2}{2^n}.$$

Proof. Proof is similar to the proof of $F_{B_1}^{\text{EDM}}$ and $F_{B_2}^{\text{EDM}}$ is deferred to Appendix A.5.

In Appendix A.6, we have shown an RBT_k attack with $O(2^{n/2})$ queries for the $F_{B_3}^{\text{EDM}}$ construction.

4.3.3 CDY and CDY_k security of $F_{B_3}^{\text{EDM}}$

Theorem 6. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Assume that the context selector S does not query $F_{B_3}^{\text{EDM}}$, H , or the ideal cipher E . For any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies

$$\text{Adv}_{F_{B_3}^{\text{EDM}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p) + \frac{2q_p q_h}{2^n} + \frac{2q_p^2}{2^n} + \epsilon_0,$$

where ϵ_0 is the probability that S outputs the challenge tag 0^n .

Proof. The proof is similar as of $F_{B_1}^{\text{EDM}}$ and $F_{B_2}^{\text{EDM}}$. It is deferred to Appendix A.7.

Therefore, $F_{B_3}^{\text{EDM}}$ achieves CDY security against $O(2^{n/2})$ queries. By Lemma 1, it further satisfies CDY_k security for the same query bound.

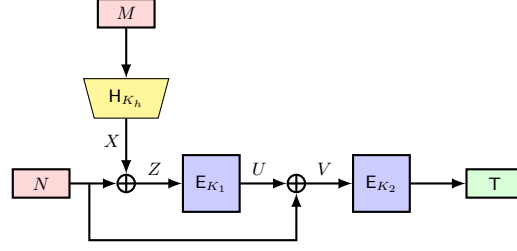
A matching attack on CDY_k is described in Appendix A.8. The attack establishes that the above bound is tight for both CDY_k and CDY security of $F_{B_3}^{\text{EDM}}$.

4.4 $F_{B_4}^{\text{EDM}}$

Let E be a block cipher and H be a hash function, the function $F_{B_4}^{\text{EDM}}$ is defined as

$$F_{B_4}^{\text{EDM}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N \oplus H(K_h, M)) \oplus N).$$

and Chen et al. [10] prove that it achieves $3n/4$ -bit MAC security against nonce-respecting adversaries. Recently, Choi et al. [11] proved the tightness of this bound assuming that the underlying hash function is xor-collision resistant.

Fig. 9: $F_{B_4}^{\text{EDM}}$

4.4.1 CMT and CMT_k Security of $F_{B_4}^{\text{EDM}}$ Here, we present a key-committing attack against the $F_{B_4}^{\text{EDM}}$ construction. The adversary begins by selecting a hash key K_h and a message M , then computes $X = H(K_h, M)$. Next, the adversary chooses two distinct keys, K and K' , and derives the corresponding nonce values N and N' as $E_K^{-1}(X) \oplus X$ and $E_{K'}^{-1}(X) \oplus X$, respectively. Finally, the adversary outputs the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) . The complete attack procedure is detailed in Figure 10 and the analysis is described in Appendix A.9.

- 1: Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$ and compute $X = H(K_h, M)$;
- 2: Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
- 3: Compute $N = E_K^{-1}(X) \oplus X$, $N' = E_{K'}^{-1}(X) \oplus X$;
- 4: Output the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) ;

Fig. 10: CMT_k adversary against $F_{B_4}^{\text{EDM}}$

Therefore, the construction $F_{B_4}^{\text{EDM}}$ is not CMT_k secure, and by Lemma 1, we can conclude that it is also not CMT secure.

4.4.2 RBT_k security of $F_{B_4}^{\text{EDM}}$

Theorem 7. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. For any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key-robustness advantage satisfies

$$\text{Adv}_{F_{B_4}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2 q_h) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^3) + \frac{6q_p^2}{2^n} + \frac{2q_p^4}{2^n} + \frac{2q_p^3 q_h}{2^n} + \frac{2q_p^2 q_h^2}{2^n}.$$

Proof. Let $\mathcal{A}$ output $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that

1. $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$, and
2. $F_{B_4}^{\text{EDM}}(K_1, K_2, K_h, N, M) = F_{B_4}^{\text{EDM}}(K'_1, K'_2, K'_h, N, M)$.

Define

$$T := E_{K_2}(E_{K_1}(N \oplus H(K_h, M)) \oplus N), \quad T' := E_{K'_2}(E_{K'_1}(N \oplus H(K'_h, M)) \oplus N).$$

Thus the adversary succeeds only if $T = T'$. We analyze this event in two main cases.

Case 1: $K_2 = K'_2$. Then

$$\begin{aligned} E_{K_1}(N \oplus H(K_h, M)) \oplus N &= E_{K'_1}(N \oplus H(K'_h, M)) \oplus N \\ \implies U := E_{K_1}(N \oplus X) &= E_{K'_1}(N \oplus X') =: U', \end{aligned}$$

where

$$X = H(K_h, M), \quad X' = H(K'_h, M), \quad Z = N \oplus X, \quad Z' = N \oplus X'.$$

We distinguish subcases according to the ordering of the relevant hash and ideal-cipher queries.

- 1a. If $H(K_h, M) = H(K'_h, M)$ but $K_h \neq K'_h$, then this is a hash collision event. Its probability is upper bounded by $\text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$.
- 1b. Otherwise, suppose the last among the two relevant ideal-cipher queries is a forward query. Then the probability that $U = U'$ is at most $2q_p^2/2^n$.
- 1c. Suppose the last relevant query is inverse. W.l.o.g., let (K'_1, U) be the final query. Then $\mathcal{A}$ succeeds if

$$E_{K'_1}^{-1}(U) = E_{K_1}^{-1}(U) \oplus H(K_h, M) \oplus H(K'_h, M).$$

- 1c1. If $E_{K'_1}^{-1}(U)$ occurs after both relevant hash queries, then the success probability is at most $2q_p^2q_h^2/2^n$.
- 1c2. If one of the hash queries (w.l.o.g. (K'_h, M, X')) occurs after the other three relevant queries, then success requires solving a preimage problem for H with target set

$$\{E_{K'_1}^{-1}(U) \oplus E_{K_1}^{-1}(U) \oplus H(K_h, M)\},$$

which has cardinality at most $q_p^2q_h$. So, the success probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2q_h)$.

Case 2: $K_2 \neq K'_2$. Now the relevant ideal-cipher queries involve distinct keys. W.l.o.g. assume (K'_2, V', T') is queried after (K_2, V, T) . For success, we must have

$$\begin{aligned} E_{K_2}(V) &= T = T' = E_{K'_2}(V'), \\ V \oplus U &= V' \oplus U' = N, \\ E_{K_1}^{-1}(U) &= N \oplus H(K_h, M), \\ E_{K'_1}^{-1}(U') &= N \oplus H(K'_h, M). \end{aligned}$$

We again distinguish subcases.

- 2a. If the forward query on (K'_2, V') occurs after (K_2, V, T) , then the collision probability is at most $2q_p^2/2^n$.
- 2b. If the inverse query on (K'_2, T) occurs after (K_2, V, T) and satisfies

$$E_{K'_2}^{-1}(T) = V = V' = E_{K_2}^{-1}(T),$$

then, by ideal-cipher randomness, this event occurs with probability at most $2q_p^2/2^n$.

- 2c. If the inverse query on (K'_2, T) occurs after (K_1, Z, U) , (K'_1, Z', U') , and (K_2, V, T) , or symmetrically a forward query on (K_1, Z') occurs after the other three, then the probability that all required equalities hold is at most $2q_p^4/2^n$.
- 2d. Otherwise, the last query among (K_1, Z, U) , (K'_1, Z', U') , (K_2, V, T) , (K'_2, V', T') is an inverse query on one of the first-layer keys. W.l.o.g. let it be (K'_1, U') .

Then $\mathcal{A}$ wins if

$$E_{K'_1}^{-1}(U') = V \oplus U \oplus H(K_h, M).$$

If the inverse query occurs after $H(K_h, M)$, the success probability is at most $2q_p^3q_h/2^n$.

If the hash query occurs after the inverse query, then success requires a preimage for H with the target set

$$\{E_{K'_1}^{-1}(U') \oplus V \oplus U\},$$

whose size is at most q_p^3 . Hence, the probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^3)$.

Summing the bounds from all subcases yields the claimed result.

We have also demonstrated a matching attack for RBT_k on $F_{B_4}^{\text{EDM}}$ in Appendix A.10.

4.4.3 CDY Security of $F_{B_4}^{\text{EDM}}$

Theorem 8. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access $F_{B_4}^{\text{EDM}}$, H , or the ideal cipher E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to E , with $q_p \leq 2^{n-1}$, the context-discovery advantage satisfies

$$\text{Adv}_{F_{B_4}^{\text{EDM}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2) + \frac{2q_p}{2^n} + \frac{2q_p^2 q_h}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a tuple (K_1, K_2, K_h, N, M) such that

$$F_{B_4}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T.$$

The adversary may query both the forward and inverse directions of the ideal cipher, as well as the hash oracle. We bound the success probability by considering the following cases.

Case 1. Suppose the ideal-cipher queries involving (K_1, Z, U) and (K_2, V, T) satisfy $Z = V$. Then

$$Z = V \implies N \oplus H(K_h, M) = N \oplus U \implies H(K_h, M) = U = T.$$

Thus, the adversary must find a preimage of T under H , occurs with probability at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2. Suppose the adversary makes forward ideal-cipher queries under K_2 and attempts to obtain T as an output. Since E is an ideal cipher, the probability of getting T is at most $2q_p/2^n$.

Case 3. Suppose the adversary makes inverse ideal-cipher queries involving T . Then $\mathcal{A}$ succeeds only if

$$\begin{aligned} E_{K_1}(N \oplus H(K_h, M)) \oplus N &= E_{K_2}^{-1}(T) \\ \implies E_{K_2}^{-1}(T) &= E_{K_1}(Z) \oplus Z \oplus H(K_h, M). \end{aligned}$$

If one of the relevant ideal-cipher queries is the final query among the three relevant queries, then by randomness of the ideal cipher, the probability that the above equality holds is at most $2q_p^2 q_h / 2^n$.

If instead the hash query is the final query, then $\mathcal{A}$ must find a preimage of H for the target set

$$\{E_{K_2}^{-1}(T) \oplus E_{K_1}(Z) \oplus Z\},$$

whose cardinality is at most q_p^2 . Hence, the success probability in this case is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2)$.

Summing the bounds from all three cases yields the claimed result.

A matching attack on CDY is described in Appendix A.11.

4.4.4 CDY_k security of $F_{B_4}^{\text{EDM}}$

Theorem 9. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access any of $F_{B_4}^{\text{EDM}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:

$$\text{Adv}_{F_{B_4}^{\text{EDM}}}^{\text{CDY}_k}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p) + \frac{2q_p}{2^n} + \frac{4q_p^2}{2^n} + \frac{2q_p q_h}{2^n}.$$

Proof. Let (N, M, T) denote the challenge tuple. Let the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h) such that

$$F_{B_4}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T.$$

The adversary is allowed to make queries to the ideal cipher (both forward and inverse) as well as to the hash oracle. We analyze the success probability of $\mathcal{A}$ by considering the following cases:

Case 1: Let the ideal cipher queries involving (K_1, Z, U) and (K_2, V, T) be exactly the same. Then we have

$$Z = V \implies N \oplus H(K_h, M) = N \oplus U \implies H(K_h, M) = U = T.$$

Thus, the adversary must find a preimage of T under H , occurs with probability at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: The adversary makes forward ideal cipher queries involving K_2 and attempts to find a collision between one of the outputs and the challenge tag T . Due to the randomness of the ideal cipher, the probability of a successful collision is at most $2q_p/2^n$.

Case 3: The adversary makes inverse ideal cipher queries involving T . In this case, $\mathcal{A}$ succeeds if and only if

$$E_{K_1}(Z) \oplus N = E_{K_2}^{-1}(T), \quad \text{where } Z = N \oplus X.$$

There are three subcases based on the order and type of the queries:

- If the query (K_1, Z, U) is made before the inverse query (K_2, V, T) , then the success probability is at most $2q_p^2/2^n$.
- If the forward query (K_1, Z, U) is made after the inverse query (K_2, V, T) , then the adversary must satisfy $E_{K_1}(N \oplus X) \oplus N = V$, which occurs with probability at most $2q_p^2/2^n$.
- If the inverse query (K_1, Z, U) is made after the inverse query (K_2, V, T) , then the adversary must satisfy $E_{K_1}^{-1}(U) = N \oplus H(K_h, M)$.

If the inverse query on (K_1, U) occurs after the hash query on (K_h, M) , then, by the ideal-cipher randomness, $\mathcal{A}$ succeeds with probability at most $2q_p q_h/2^n$.

On the other hand, if the hash query on (K_h, M) occurs after the inverse ideal-cipher query, then $\mathcal{A}$ succeeds only if it finds a preimage under H for the target set $\{E_{K_1}^{-1}(U) \oplus N\}$.

Clearly, this target set has cardinality at most q_p . Hence, in this case, $\mathcal{A}$'s success probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p)$.

Therefore, combining all the cases, the result follows.

A matching attack on CDY_k is described in Appendix A.12.

4.5 $F_{B_5}^{\text{EDM}}$

Let E be a block cipher and H be a hash function. The function $F_{B_5}^{\text{EDM}}$ is defined as

$$F_{B_5}^{\text{EDM}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N \oplus H(K_h, M)) \oplus N) \oplus H(K_h, M),$$

and Chen et al. [10] prove that it achieves $3n/4$ -bit MAC security against nonce-respecting adversaries. Recently, Choi et al. [11] showed the tightness of this bound assuming that the underlying hash function is xor-collision resistant.

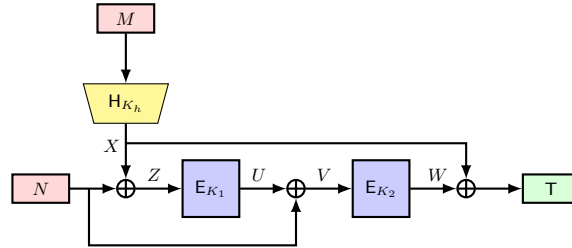


Fig. 11: $F_{B_5}^{\text{EDM}}$

4.5.1 CMT and CMT_k Security of $F_{B_5}^{\text{EDM}}$ The key-committing attack against $F_{B_5}^{\text{EDM}}$ follows the same procedure as the attack on $F_{B_4}^{\text{EDM}}$. For completeness, we provide the full attack algorithm in Figure 12 and the analysis is described in Appendix A.13.

- 1: Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$ and compute $X = H(K_h, M)$;
- 2: Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
- 3: Compute $N = E_K^{-1}(X) \oplus X$ and $N' = E_{K'}^{-1}(X) \oplus X$;
- 4: Output the tuples (K, K, K_h, N, M) and (K', K', K_h, N', M) ;

Fig. 12: CMT_k adversary against $F_{B_5}^{\text{EDM}}$

Therefore, the construction $F_{B_5}^{\text{EDM}}$ is not CMT_k secure, and by Lemma 1, it is also not CMT secure.

4.5.2 RBT_k security of $F_{B_5}^{\text{EDM}}$

Theorem 10. Let $H: \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key-robustness advantage satisfies

$$\text{Adv}_{F_{B_5}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_{F_{B_4}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A}') + \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2 q_h) + \frac{2q_p^2 q_h^2}{2^n} + \frac{2q_p^2}{2^n} + \frac{2q_p^4}{2^n} + \frac{2q_p^3 q_h}{2^n}.$$

Proof. The detailed proof is deferred to appendix A.14.

In Appendix A.15, we have shown an RBT_k attack with $O(2^{n/2})$ queries for the $F_{B_5}^{\text{EDM}}$ construction.

4.5.3 CDY and CDY_k security of $F_{B_5}^{\text{EDM}}$

Theorem 11. Let $H: \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access $F_{B_5}^{\text{EDM}}$, H , or E . Then for any adversary $\mathcal{A}$ making at most q_h hash queries and q_p ideal-cipher queries, with $q_p \leq 2^{n-1}$,

$$\text{Adv}_{F_{B_5}^{\text{EDM}}}^{\text{CDY}}(\mathcal{A}; S) \leq \epsilon_0 + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p) + \frac{2q_p q_h}{2^n} + \frac{2q_p^2}{2^n},$$

where ϵ_0 is the probability that S outputs 0^n .

Proof. Let T denote the challenge tag. Suppose $\mathcal{A}$ outputs (K_1, K_2, K_h, N, M) such that

$$F_{B_5}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T.$$

The adversary may query both directions of E and the hash oracle.

If $T = 0^n$, the adversary can always construct a valid context: choose any K_h, M and compute $X = H(K_h, M)$; choose any K and set $N = E_K^{-1}(X) \oplus X$. Then

$$F_{B_5}^{\text{EDM}}(K, K, K_h, N, M) = E_K(E_K(N \oplus X) \oplus N) \oplus X = E_K(X \oplus N) \oplus X = X \oplus X = 0^n.$$

Hence the success probability in this case is at most ϵ_0 .

This also shows that if $(K_1, N) = (K_2, V)$ for a valid context, then necessarily $T = 0^n$. Thus in the following assume $T \neq 0^n$ and $(K_1, N) \neq (K_2, V)$. Writing $V = E_{K_1}(N \oplus H(K_h, M)) \oplus N$, success requires $E_{K_2}(V) \oplus H(K_h, M) = T$.

Case 1: If the hash query $H(K_h, M)$ occurs after the ideal-cipher query (K_2, V, W) , then success requires finding a preimage of H for the set $\{W \oplus T\}$, whose size is at most q_p . Hence the probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p)$.

Case 2: If a forward ideal-cipher query on (K_2, V) occurs after the hash query, then by randomness of E the success probability is at most $2q_p q_h / 2^n$.

Case 3: If an inverse ideal-cipher query on (K_2, W) occurs after the hash query, then success requires

$$E_{K_2}^{-1}(W) = E_{K_1}(Z) \oplus Z \oplus T \oplus W,$$

which holds with probability at most $2q_p^2 / 2^n$.

Summing all cases yields the claimed bound.

Therefore, $F_{B_5}^{\text{EDM}}$ achieves CDY security against $O(2^{n/2})$ queries. By Lemma 1, it further satisfies CDY_k security for the same query bound.

A matching attack on CDY_k is described in Appendix A.16. The attack establishes that the above bound is tight for both CDY_k and CDY security of $F_{B_5}^{\text{EDM}}$.

5 Analysis of Nonce-based MACs (based on EDMD)

Chen et al. [10] introduced nonce-based PRFs based on the Encrypted Davies Meyer Dual (EDMD) construction, denoted as F^{EDMD} . Let $\kappa, \kappa_h, n \in \mathbb{N}$, $E: \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher and $H: \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function. They proposed a generic construction $F_{B_x}^{\text{EDMD}}: \{0, 1\}^{2\kappa} \times \{0, 1\}^{\kappa_h} \times \{0, 1\}^n \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ defined as

$$F_{B_x}^{\text{EDMD}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N \oplus b_1 \cdot H_{K_h}(M)) \oplus b_1 \cdot H_{K_h}(M)) \oplus E_{K_1}(N \oplus b_1 \cdot H_{K_h}(M)) \oplus b_3 \cdot H_{K_h}(M),$$

where $B_x \in \{B_0, \dots, B_7\}$ is the 3-bit representation of x , denoted as (b_1, b_2, b_3) . Among the eight possible constructions, the authors conclude that only $F_{B_1}^{\text{EDMD}}$ achieves beyond-birthday-bound MAC security against nonce-respecting adversaries, while $F_{B_2}^{\text{EDMD}}$ lacks a security analysis in this regard.

5.1 $F_{B_1}^{\text{EDMD}}$

The function $F_{B_1}^{\text{EDMD}}$ is a Wegman Carter construction that uses EDMD as its underlying PRF. It is optimally n -bit secure against nonce-respecting adversaries and totally broken when the nonce is reused. Let E be a block cipher and H be a hash function, $F_{B_1}^{\text{EDMD}}$ is defined as

$$F_{B_1}^{\text{EDMD}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_2}(E_{K_1}(N)) \oplus E_{K_1}(N) \oplus H(K_h, M).$$

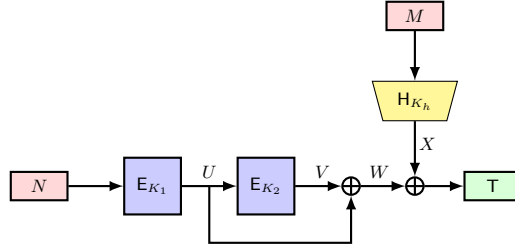


Fig. 13: $F_{B_1}^{\text{EDMD}}$

5.1.1 CMT and CMT_k security of $F_{B_1}^{\text{EDMD}}$ Here we present a key-committing attack against $F_{B_1}^{\text{EDMD}}$ construction. The adversary selects two distinct keys, K_1 and K'_1 , and computes $N = E_{K_1}^{-1}(0^n)$ and $N' = E_{K'_1}^{-1}(0^n)$. Next, the adversary selects another key K_2 , a hash key K_h , and a message M . Finally, it outputs the tuples (K_1, K_2, K_h, N, M) and (K'_1, K_2, K_h, N', M) . The complete attack description is provided in Figure 14.

- 1 : Select $K_1 \in \{0, 1\}^\kappa$ and compute $N = E_{K_1}^{-1}(0^n)$;
- 2 : Select $K'_1 \in \{0, 1\}^\kappa$ such that $K'_1 \neq K_1$ and compute $N' = E_{K'_1}^{-1}(0^n)$;
- 3 : Select $K_2 \in \{0, 1\}^\kappa$, $K_h \in \{0, 1\}^{\kappa_h}$ and $M \in \{0, 1\}^*$;
- 4 : Output the tuples (K_1, K_2, K_h, N, M) and (K'_1, K_2, K_h, N', M) ;

Fig. 14: CMT_k adversary against $F_{B_1}^{\text{EDMD}}$

Analysis of the Attack: For the first output tuple (K_1, K_2, K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_1}^{\text{EDMD}}(K_1, K_2, K_h, N, M) &= E_{K_2}(E_{K_1}(N)) \oplus E_{K_1}(N) \oplus H(K_h, M), \\ &= E_{K_2}(0^n) \oplus H(K_h, M), \text{ since } E_{K_1}(N) = 0^n. \end{aligned}$$

For the second output tuple (K'_1, K_2, K_h, N', M) , the tag is computed as:

$$\begin{aligned} F_{B_1}^{\text{EDMD}}(K'_1, K_2, K_h, N', M) &= E_{K_2}(E_{K'_1}(N')) \oplus E_{K'_1}(N') \oplus H(K_h, M), \\ &= E_{K_2}(0^n) \oplus H(K_h, M), \text{ since } E_{K'_1}(N') = 0^n. \end{aligned}$$

From the above two equations, it is clear that both tuples produce the same tag $T = E_{K_2}(0^n) \oplus H(K_h, M)$. Thus, the algorithm presented in Figure 14 leads to a CMT_k attack on $F_{B_1}^{\text{EDMD}}$ with a success probability of 1.

Therefore, the construction $F_{B_1}^{\text{EDMD}}$ is not CMT_k secure and by Lemma 1, we can conclude that $F_{B_1}^{\text{EDMD}}$ is not CMT secure also.

5.1.2 RBT_k Security of $F_{B_1}^{\text{EDMD}}$

Theorem 12. *Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key robustness advantage satisfies*

$$\text{Adv}_{F_{B_1}^{\text{EDMD}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2 q_h) + \frac{4q_p^2}{2^n} + \frac{2q_p^2 q_h^2}{2^n}.$$

Proof. Suppose $\mathcal{A}$ outputs $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that

$$(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h).$$

Define $T := E_{K_2}(E_{K_1}(N)) \oplus E_{K_1}(N) \oplus H(K_h, M)$, and $T' := E_{K'_2}(E_{K'_1}(N)) \oplus E_{K'_1}(N) \oplus H(K'_h, M)$. The adversary wins if and only if $T = T'$. We analyze the success probability by considering the following cases.

Case 1: $K_h = K'_h$. In this case, if $K_2 = K'_2$, then $\mathcal{A}$ must satisfy $E_{K_1}(N) = E_{K'_1}(N)$ with $K_1 \neq K'_1$. This occurs with probability at most $2q_p^2/2^n$. On the other hand, if $K_2 \neq K'_2$, then $\mathcal{A}$ wins if $E_{K_2}(U) \oplus U = E_{K'_2}(U') \oplus U'$. Clearly, we can then construct an adversary $\mathcal{B}$ against the collision resistance of the Davies–

Meyer construction DM. Thus $\mathcal{A}$'s success probability is upper bounded by $\text{Adv}_{\text{DM}}^{\text{coll}}(\mathcal{B}) \stackrel{(1)}{\leq} 2q_p^2/2^n$, where inequality (1) follows from Lemma 2.

Case 2: $K_h \neq K'_h$. If the adversary finds $K_h \neq K'_h$ such that $H(K_h, M) = H(K'_h, M)$, then the adversary's success probability is upper bounded by the hash collision probability $\text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$. Otherwise, if $H(K_h, M) \neq H(K'_h, M)$ and the adversary $\mathcal{A}$ wins, then

$$E_{K_2}(U) \oplus U \oplus H(K_h, M) = E_{K'_2}(U') \oplus U' \oplus H(K'_h, M).$$

If either of the hash queries (w.l.o.g. (K'_h, M)) is the last among these four relevant queries, then $\mathcal{A}$ wins only if it can find a preimage against H for the target set $\{E_{K_2}(U) \oplus U \oplus H(K_h, M) \oplus E_{K'_2}(U') \oplus U'\}$. As the cardinality of the target set is upper bounded by $q_p^2 q_h$, $\mathcal{A}$'s success probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2 q_h)$.

If either of the ideal cipher queries is the last among the four relevant queries, then $\mathcal{A}$'s success probability is upper bounded by $2q_p^2 q_h^2/2^n$.

Combining all cases, the theorem follows.

We have also demonstrated a matching attack for RBT_k on $F_{B_1}^{\text{EDMD}}$ in Appendix B.1.

5.1.3 CDY and CDY_k security of $F_{B_1}^{\text{EDMD}}$

Theorem 13. *Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context selector S does not access the construction $F_{B_1}^{\text{EDMD}}$, nor the underlying primitives H or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies*

$$\text{Adv}_{F_{B_1}^{\text{EDMD}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p) + \frac{4q_p q_h}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose $\mathcal{A}$ outputs (K_1, K_2, K_h, N, M) such that

$$F_{B_1}^{\text{EDMD}}(K_1, K_2, K_h, N, M) = T.$$

The adversary may query both directions of E and the hash oracle. $\mathcal{A}$ will be successful if

$$H(K_h, M) \oplus E_{K_2}(U) \oplus U = T.$$

Case 1. If the hash query $H(K_h, M)$ occurs after the ideal-cipher query (K_2, U, V) , then success requires finding a preimage of H for the set $\{E_{K_2}(U) \oplus U \oplus T\}$, whose size is at most q_p . Hence the probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p)$.

Case 2. If a forward ideal-cipher query on (K_2, U) occurs after the hash query, then by randomness of E the success probability is at most $2q_p q_h/2^n$.

Case 3. If an inverse ideal-cipher query on (K_2, V) occurs after the hash query, then success requires

$$E_{K_2}^{-1}(V) = T \oplus H(K_h) \oplus V,$$

which holds with probability at most $2q_p q_h / 2^n$.
Combining all the cases, we have the theorem.

Therefore, by Lemma 1 and Theorem 13, we can also conclude that $F_{B_1}^{\text{EDMD}}$ achieve the same CDY_k security as CDY security.

We will show the tightness of this bound by showing a tight matching attack in appendix B.2.

6 Analysis of Nonce-based MACs (based on SoP)

Chen et al. [10] introduced nonce-based PRFs based on the Sum of Permutation (SoP) construction, denoted as F^{SoP} . Let $\kappa, \kappa_h, n \in \mathbb{N}$, $E: \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher, and $H: \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function. They proposed a generic construction

$$F_{B_x}^{\text{SoP}}: \{0, 1\}^{2\kappa} \times \{0, 1\}^{\kappa_h} \times \{0, 1\}^n \times \{0, 1\}^* \rightarrow \{0, 1\}^n,$$

defined as

$$\begin{aligned} F_{B_x}^{\text{SoP}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) \\ = E_{K_1}(N \oplus b_1 \cdot H(K_h, M)) \oplus E_{K_2}(N \oplus b_2 \cdot H(K_h, M)) \oplus b_3 \cdot H(K_h, M), \end{aligned}$$

where $B_x \in \{B_0, \dots, B_7\}$ is the 3-bit representation of x , denoted as (b_1, b_2, b_3) .

The authors showed that among the eight possible constructions, three ($F_{B_1}^{\text{SoP}}$, $F_{B_2}^{\text{SoP}}$, and $F_{B_3}^{\text{SoP}}$) achieve beyond-birthday-bound MAC security against nonce-respecting adversaries.

6.1 $F_{B_1}^{\text{SoP}}$

The function $F_{B_1}^{\text{SoP}}$ is a Wegman–Carter MAC construction that uses the Sum of Permutations (SoP) as its underlying PRF. It achieves optimal n -bit security against nonce-respecting adversaries, but becomes completely insecure if the nonce is ever reused.

Let E be a block cipher and H be a hash function. The construction $F_{B_1}^{\text{SoP}}$ is defined as:

$$F_{B_1}^{\text{SoP}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_1}(N) \oplus E_{K_2}(N) \oplus H(K_h, M).$$

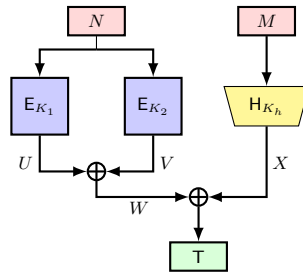


Fig. 15: $F_{B_1}^{\text{SoP}}$

6.1.1 CMT, CMT_k and RBT_k Security of $F_{B_1}^{\text{SoP}}$ We present a key-robustness attack against the $F_{B_1}^{\text{SoP}}$ construction. The adversary selects a hash key K_h , a message M , and a nonce N . Then, it chooses two distinct keys K and K' and outputs the tuples (K, K', K_h, N, M) and (K', K, K_h, N, M) . The complete attack is shown in Figure 16.

1 : Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$ and $N \in \{0, 1\}^n$;
 2 : Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
 3 : Output the tuple $((K, K, K_h), (K', K', K_h), N, M)$;

Fig. 16: RBT_k adversary against $F_{B_1}^{\text{SoP}}$

Analysis of the Attack: For any (K, K, K_h, N, M) , we have $F_{B_1}^{\text{SoP}}(K, K, K_h, N, M) = H(K_h, M)$. Therefore, the adversary described in Figure 16 constitutes a successful RBT_k attack on $F_{B_1}^{\text{SoP}}$ with success probability 1.

Therefore, the construction $F_{B_1}^{\text{SoP}}$ is not RBT_k secure and by Lemma 1 we can also conclude that it is not CMT_k and CMT secure.

6.1.2 CDY and CDY_k Security of $F_{B_1}^{\text{SoP}}$

Theorem 14. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context selector S does not access any of $F_{B_1}^{\text{SoP}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:

$$\text{Adv}_{F_{B_1}^{\text{SoP}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2) + \frac{2q_p^2 q_h}{2^n} + \frac{2q_p^2}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h, N, M) such that

$$F_{B_1}^{\text{SoP}}(K_1, K_2, K_h, N, M) = T.$$

The adversary is allowed to make queries to the ideal cipher (both forward and inverse) and to the hash oracle. We analyze the success probability of $\mathcal{A}$ by considering the following cases:

Case 1: Suppose there exist ideal cipher queries involving (K_1, N, U) and (K_2, N, V) such that $E_{K_1}(N) = U$ and $E_{K_2}(N) = V$, and $U = V$. Then,

$$F_{B_1}^{\text{SoP}}(K_1, K_2, K_h, N, M) = H(K_h, M).$$

Hence, the adversary succeeds only if it find preimage for T , which occurs with probability at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: Otherwise, $\mathcal{A}$ will be successful if

$$E_{K_1}(N) \oplus E_{K_2}(N) \oplus H(K_h, M) = T$$

Without loss of generality, assume the ideal cipher query involving (K_1, N, U) occurs before the one involving (K_2, N, V) .

- The hash query on (K_h, M) is the final query among the three relevant queries. Then $\mathcal{A}$ will be successful if it can find a preimage for the target set $\{E_{K_1}(N) \oplus E_{K_2}(N)\}$ against H . Clearly, cardinality of the target set is at most q_p^2 . Hence, $\mathcal{A}$'s success probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2)$.
- If the final query is a forward ideal cipher query (w.l.o.g $E_{K_2}(N)$), then for the adversary to succeed, we must have

$$E_{K_2}(N) = E_{K_1}(N) \oplus H(K_h, M) \oplus T.$$

Then the success probability is at most $2q_p^2 q_h / 2^n$.

- If the final query is an inverse ideal cipher query (w.l.o.g $E_{K_2}^{-1}(V)$) then the adversary must satisfy

$$E_{K_2}^{-1}(E_{K_1}(N) \oplus H(K_h, M) \oplus T) = N.$$

As we have assumed that (K_1, N, U) query is done before the query (K_2, N, V) query, so target value of N for the $E_{K_2}^{-1}$ is at most q_p . Moreover, at most q_p ideal cipher query assumption gives that the success probability in this case is at most $2q_p^2 / 2^n$.

Hence, combining all cases, we obtain the bound as stated.

Therefore, $F_{B_1}^{\text{SoP}}$ achieves CDY security against $O(2^{n/3})$ queries. By Lemma 1, it further satisfies CDY_k security for the same query bound.

A matching attack on CDY_k is described in Appendix A.3. The attack establishes that the above bound is tight for both CDY_k and CDY security of $F_{B_2}^{\text{EDM}}$.

6.2 $F_{B_2}^{\text{SoP}}$

The function $F_{B_2}^{\text{SoP}}$ is defined as

$$F_{B_2}^{\text{SoP}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_1}(N) \oplus E_{K_2}(N \oplus H(K_h, M)),$$

where E is a block cipher and H is a hash function. Chen et al. [10] proved that $F_{B_2}^{\text{SoP}}$ achieves $3n/4$ -bit MAC security against nonce-respecting adversaries. More recently, Choi et al. [11] improved this result by showing an optimal bound under the assumption that the underlying hash function is multi-xor-collision resistant. Their analysis also proves that the bound is tight.

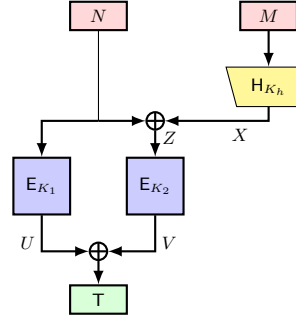


Fig. 17: $F_{B_2}^{\text{SoP}}$

6.2.1 CMT and CMT_k Security of $F_{B_2}^{\text{SoP}}$ Here we present a key-committing attack against the $F_{B_2}^{\text{SoP}}$ construction. The adversary selects a hash key K_h , a message M , a nonce N , and computes $X = H(K_h, M)$. Next, the adversary selects two distinct keys, K and K' , and outputs the tuples (K, K', K_h, N, M) and $(K', K, K_h, N \oplus X, M)$. The complete attack description is provided in Figure 18 and the analysis is described in Appendix C.2.

- 1 : Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$, $N \in \{0, 1\}^n$ and compute $X = H(K_h, M)$;
- 2 : Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
- 3 : Output the tuples (K, K', K_h, N, M) and $(K', K, K_h, N \oplus X, M)$;

Fig. 18: CMT_k adversary against $F_{B_2}^{\text{SoP}}$

Therefore, the construction $F_{B_2}^{\text{SoP}}$ is not CMT_k secure and from the security relations as stated in Lemma 1, we can conclude that $F_{B_2}^{\text{SoP}}$ is not CMT secure also.

6.2.2 RBT_k security of $F_{B_2}^{\text{SoP}}$

Theorem 15. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key robustness advantage satisfies:

$$\text{Adv}_{F_{B_2}^{\text{SoP}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2 q_h) + \frac{4q_p^2}{2^n} + \frac{4q_p^4}{2^n} + \frac{4q_p^2 q_h^2}{2^n}.$$

Proof. Detailed proof is deferred to appendix C.3.

We have also demonstrated a matching attack for RBT_k on $F_{B_2}^{\text{SoP}}$ in Appendix C.4.

6.2.3 CDY security of $F_{B_2}^{\text{SoP}}$

Theorem 16. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access any of $F_{B_2}^{\text{SoP}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:

$$\text{Adv}_{F_{B_2}^{\text{SoP}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)_{\epsilon_0} + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2) + \frac{2q_p^2 q_h}{2^n} + \frac{2q_p^2}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h, N, M) such that

$$F_{B_2}^{\text{SoP}}(K_1, K_2, K_h, N, M) = T.$$

That is,

$$T = E_{K_1}(N) \oplus E_{K_2}(N \oplus H(K_h, M)).$$

The adversary is allowed to query the ideal cipher (forward and inverse) and the hash oracle. We analyze the success probability by considering the following cases:

Case 1: Collision in input values. Suppose the inputs to both cipher calls are equal, i.e.,

$$N = N \oplus H(K_h, M) \Rightarrow H(K_h, M) = 0^n.$$

Hence, in this case adversary will be successful only if the challenge tag $T = 0^n$. The probability that the adversary queries $H(K_h, M) = 0^n$ in at most q_h queries to the hash oracle is at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$. So in this case adversary will be successful with probability at most $\epsilon_0 \cdot \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: No collision in inputs. That is, $H(K_h, M) \neq 0^n$. Let $Z = N \oplus H(K_h, M)$. $\mathcal{A}$ will be successful if $E_{K_1}(N) \oplus E_{K_2}(N \oplus H(K_h, M)) = T$.

- **Case 2a:** The final query among these two relevant queries is a forward ideal cipher query. Then $\mathcal{A}$ will be successful with probability at most $2q_p^2/2^n$.
- **Case 2b:** The final query among these two relevant queries is an inverse ideal cipher query (w.l.o.g let $E_{K_2}^{-1}(V)$). Then, $\mathcal{A}$ will be successful if

$$E_{K_2}^{-1}(E_{K_1}(N) \oplus T) = N \oplus H(K_h, M).$$

If the inverse query occurs after the hash query then $\mathcal{A}$'s success probability is upper bounded by $2q_p^2 q_h / 2^n$. If the hash query occurs after the inverse query then the probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2)$.

Combining all three cases, the result follows.

A matching attack on CDY for $F_{B_2}^{\text{SoP}}$ is described in Appendix C.5.

6.2.4 CDY_k Security of $F_{B_2}^{\text{SoP}}$

Theorem 17. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access any of $F_{B_2}^{\text{SoP}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:

$$\text{Adv}_{F_{B_2}^{\text{SoP}}}^{\text{CDY}_k}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p) + \frac{2q_p^2}{2^n}.$$

Proof. Let (N, M, T) denote the challenge tuple. Let the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h) such that

$$F_{B_2}^{\text{SoP}}(K_1, K_2, K_h, N, M) = T.$$

The adversary is allowed to make queries to the ideal cipher (both forward and inverse) and the hash oracle. We analyze the success probability of $\mathcal{A}$ by considering the following cases:

Case 1: Let the ideal cipher queries involve (K_1, N, U) and (K_2, Z, V) where $E_{K_1}(N) = U$ and $E_{K_2}(Z) = V$. These are equal only if $N = N \oplus X$, i.e., $X = 0^n$. That means the adversary must find K_h such that $H(K_h, M) = 0^n$. In this case $\mathcal{A}$'s success is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: Without loss of generality, assume the query involving (K_2, Z, V) occurs after the one involving (K_1, N, U) . If this query is forward, then the adversary must satisfy:

$$E_{K_2}(Z) = E_{K_1}(N) \oplus T.$$

By ideal cipher randomness, this occurs with probability at most $2q_p^2/2^n$. If the query is inverse, then the adversary must satisfy:

$$E_{K_2}^{-1}(E_{K_1}(N) \oplus T) = N \oplus H(K_h, M).$$

If the inverse ideal cipher query occurs after the hash query, then this happens with probability at most $2q_p q_h/2^n$. If the hash query occurs after the inverse query then this happens with probability at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p)$.

Therefore, combining both cases, the result follows.

A matching attack on CDY_k for $F_{B_2}^{\text{SoP}}$ is described in Appendix C.6.

6.3 $F_{B_3}^{\text{SoP}}$

The function $F_{B_3}^{\text{SoP}}$ is defined as

$$F_{B_3}^{\text{SoP}}[E_{K_1}, E_{K_2}, H_{K_h}](N, M) = E_{K_1}(N) \oplus E_{K_2}(N \oplus H(K_h, M)) \oplus H(K_h, M),$$

and Chen et al. [10] prove that it achieves $3n/4$ -bit MAC security against nonce-respecting adversaries. Recently, Choi et al. [11] improved this result (assuming that the underlying hash function is multi-xor-collision resistant), establishing an optimal security bound and proving its tightness.

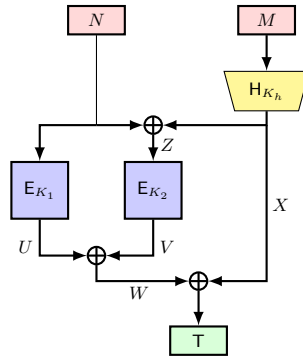


Fig. 19: $F_{B_3}^{\text{SoP}}$

6.3.1 CMT and CMT_k Security of $F_{B_3}^{\text{SoP}}$ The key-committing attack against $F_{B_3}^{\text{SoP}}$ follows the same procedure as the attack on $F_{B_2}^{\text{SoP}}$. For completeness, we provide the full attack algorithm in Figure 20 and the analysis is described in Appendix C.7.

- 1 : Select any $K_h \in \{0, 1\}^{\kappa_h}$, $M \in \{0, 1\}^*$, $N \in \{0, 1\}^n$ and compute $X = H(K_h, M)$;
- 2 : Select $K, K' \in \{0, 1\}^\kappa$ such that $K \neq K'$;
- 3 : Output the tuples (K, K', K_h, N, M) and $(K', K, K_h, N \oplus X, M)$;

Fig. 20: CMT_k adversary against $F_{B_3}^{\text{SoP}}$

Therefore, the construction $F_{B_3}^{\text{SoP}}$ is not CMT_k secure and from the security relations as stated in Lemma 1, we can conclude that $F_{B_3}^{\text{SoP}}$ is not CMT secure also.

6.3.2 RBT_k security of $F_{B_3}^{\text{SoP}}$

Theorem 18. *Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the key robustness advantage satisfies:*

$$\text{Adv}_{F_{B_3}^{\text{SoP}}}^{\text{RBT}_k}(\mathcal{A}) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2) + \text{Adv}_H^{\text{CR}}(\mathcal{A}_{\text{cr}}) + \frac{8q_p^2}{2^n} + \frac{6q_p^4}{2^n} + \frac{2q_p^2 q_h}{2^n}.$$

Proof of the above theorem is deferred to Appendix C.8.

Matching Attack for RBT_k on $F_{B_3}^{\text{SoP}}$: We now demonstrate the tightness of the RBT_k bound for $F_{B_3}^{\text{SoP}}$ as established in the preceding theorem. The attack against $F_{B_3}^{\text{SoP}}$ follows the same procedure as the attack on $F_{B_2}^{\text{SoP}}$. For completeness, we provide the attack steps in the following:

- Fix a key K_h , a nonce N and a message M . Compute $X = H(K_h, M)$.
- Select $2^{n/4}$ distinct values for K_1^i , and for each $i \in [1, 2^{n/4}]$, compute $U_i = E_{K_1^i}(N)$.
- Select $2^{n/4}$ distinct values for K_2^i , and for each $i \in [1, 2^{n/4}]$, compute $V_i = E_{K_2^i}(N \oplus X)$.
- Find indices $a, b, c, d \in [1, 2^{n/4}]$ such that $U_a \oplus V_b = U_c \oplus V_d$.
- Output the tuple $((K_1^a, K_2^b, K_h), (K_1^c, K_2^d, K_h), N, M)$.

Since each U_i and V_i is sampled uniformly at random and independently, the success probability of the attack is at least $1 - e^{-1}$.

6.3.3 CDY and CDY_k Security of $F_{B_3}^{\text{SoP}}$

Theorem 19. *Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a keyed hash function. Suppose the context-selector S does not access any of $F_{B_3}^{\text{SoP}}$, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies:*

$$\text{Adv}_{F_{B_3}^{\text{SoP}}}^{\text{CDY}}(\mathcal{A}; S) \leq \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1) \cdot \epsilon_0 + \frac{2q_p^2}{2^n},$$

where ϵ_0 denotes the probability that S outputs the challenge tag 0^n .

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a valid tuple (K_1, K_2, K_h, N, M) such that

$$F_{B_3}^{\text{SoP}}(K_1, K_2, K_h, N, M) = T.$$

The adversary is allowed to make queries to the ideal cipher (both forward and inverse) and to the hash oracle. We analyze the success probability of $\mathcal{A}$ by considering the following cases:

Case 1: The adversary finds (K_1, K_2, K_h, N, M) such that

$$(K_1, N) = (K_2, N \oplus X), \quad \text{where } X = H(K_h, M).$$

This implies $X = 0^n$, so the tag must be

$$T = E_{K_1}(N) \oplus E_{K_2}(N) \oplus 0^n = 0^n.$$

Therefore, the adversary must find a hash query (K_h, M) such that $H(K_h, M) = 0^n$. Probability of finding such pair is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$. This is relevant only when $T = 0^n$, which occurs with probability ϵ_0 . Hence, in this case the probability of success is at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)\epsilon_0$.

Case 2: The adversary finds a valid tuple where $X = H(K_h, M) \neq 0^n$, and succeeds with

$$F_{B_3}^{\text{SoP}}(K_1, K_2, K_h, N, M) = E_{K_1}(N) \oplus E_{K_2}(N \oplus X) \oplus X = T.$$

This implies

$$E_{K_2}(N \oplus X) = T \oplus X \oplus E_{K_1}(N),$$

or equivalently,

$$E_{K_1}(N) \oplus N = E_{K_2}(N \oplus X) \oplus (N \oplus X) \oplus T.$$

Note that T is chosen independently by the context selector. Since all ideal cipher evaluations are independent and uniform, and the adversary makes at most q_p queries, the probability of such a collision is bounded by $2q_p^2/2^n$.

Combining both cases, the result follows.

Therefore, $F_{B_3}^{\text{SoP}}$ achieves CDY security against $O(2^{n/2})$ queries. By Lemma 1, it further satisfies CDY_k security for the same query bound.

A matching attack on CDY_k is described in Appendix C.9. The attack establishes that the above bound is tight for both CDY_k and CDY security of $F_{B_3}^{\text{SoP}}$.

7 Analysis of nEHtM

Dutta et. al. [18] have proposed a nonce-based version of EHtM [32] construction, called *nonce-based Enhanced Hash-then-Mask* (nEHtM), and gives upto $2n/3$ -bit unforgeability in faulty nonce model. For the purpose of domain separation, they consider an $(n-1)$ -bit nonce and $(n-1)$ -bit hash function. For any message M and nonce N , nEHtM is defined as follows

$$\text{nEHtM}[E_K, H_{K_h}](N, M) = E_K(0\|N) \oplus E_K(1\|(N \oplus H(K_h, M))).$$

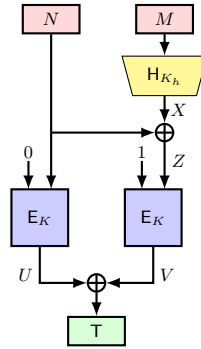


Fig. 21: nEHtM

7.1 CMT security of nEHtM

Theorem 20. Let $H : \{0, 1\}^{\kappa_h} \times \{0, 1\}^* \rightarrow \{0, 1\}^{n-1}$ be a keyed hash function. Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the commitment security advantage satisfies

$$\text{Adv}_{\text{nEHtM}}^{\text{CMT}}(\mathcal{A}) \leq \text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr}) + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2) + \frac{2q_p^4}{2^n} + \frac{2q_p^2 q_h}{2^n}.$$

Proof. Let $\mathcal{A}$ output two tuples $(K, K_h, N, M) \neq (K', K'_h, N', M')$ such that

$$\text{nEHtM}(K, K_h, N, M) = \text{nEHtM}(K', K'_h, N', M') = T.$$

Let, $U = E_K(0\|N)$, $V = E_K(1\|(N \oplus H(K_h, M)))$, $U' = E_{K'}(0\|N')$, and $V' = E_{K'}(1\|(N' \oplus H(K'_h, M')))$. Then, $T = U \oplus V = U' \oplus V'$. Let $X = H(K_h, M)$, $X' = H(K'_h, M')$, $Z = N \oplus X$, and $Z' = N' \oplus X'$. Now, we consider the following cases:

- **Case 1:** $(K, N) = (K', N')$. For the tag collision, we must have $Z = Z'$, which implies $X = X'$. Since $(K_h, M) \neq (K'_h, M')$, this requires a hash collision. Hence, the adversary's success probability is bounded by $\text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$.
- **Case 2:** $(K, N) \neq (K', N')$ and $(K, Z) \neq (K', Z')$. The adversary succeeds if

$$E_K(0\|N) \oplus E_K(1\|Z) = E_{K'}(0\|N') \oplus E_{K'}(1\|Z').$$

- If the last query among these four is forward, then by the randomness of the ideal cipher, the probability of success is bounded by $2q_p^4/2^n$.

- If the last query is an inverse query, without loss of generality, suppose that it corresponds to K' and V' . The adversary then succeeds if

$$\langle E_{K'}^{-1}(V') \rangle_{n-1} = N' \oplus X'.$$

If the inverse ideal cipher query occurs after the hash query involving (K'_h, M', X') , this probability is bounded by $2q_p^2 q_h / 2^n$. If the hash query occurs after the inverse ideal cipher query, then $\mathcal{A}$ succeeds if it can find a preimage for a set of cardinality at most q_p^2 . Then, $\mathcal{A}$'s success probability is upper bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; q_p^2)$.

Similarly, if the last inverse query corresponds to K' and U' , the bound is the same.

Combining all cases, we have the theorem.

A $2^{n/2}$ -query attack on nEHtM for RBT_k described in Appendix D.1. This also shows that nEHtM can not achieve more than $n/2$ -bit security for CMT_k and CMT .

7.2 CDY security of nEHtM

Theorem 21. *Let $H : \{0,1\}^{\kappa_h} \times \{0,1\}^* \rightarrow \{0,1\}^{n-1}$ be a keyed hash function. Suppose the context-selector S does not access any of nEHtM, H , or E . Then, for any adversary $\mathcal{A}$ making at most q_h queries to H and q_p queries to the ideal cipher E , with $q_p \leq 2^{n-1}$, the context discovery advantage satisfies*

$$\text{Adv}_{\text{nEHtM}}^{\text{CDY}}(\mathcal{A}; S) \leq 2 \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; q_p) + \frac{4q_p^2}{2^n} + \frac{4q_p q_h}{2^n}.$$

Proof. Let T denote the challenge tag. Suppose the adversary $\mathcal{A}$ outputs a valid tuple (K, K_h, N, M) such that

$$\text{nEHtM}(K, K_h, N, M) = T.$$

The adversary may query the ideal cipher (both forward and inverse) and the hash oracle. Let

$$U = E_K(0||N), \quad X = H(K_h, M), \quad Z = N \oplus X, \quad V = E_K(1||Z), \quad T = U \oplus V.$$

We analyze the success probability of $\mathcal{A}$ by considering the following cases.

Case 1: The ideal cipher query involving $(K, 0||N, U)$ occurs after the query involving $(K, 1||Z, V)$.

If $(K, 0||N, U)$ is a forward query, then the adversary succeeds if

$$E_K(0||N) = V \oplus T,$$

where $V = E(K, 1||*)$ for some previous query. Since $\mathcal{A}$ makes at most q_p ideal cipher queries, the success probability in this case is at most $2q_p^2/2^n$.

If $(K, 0||N, U)$ is an inverse query, then the adversary succeeds if

$$E_K^{-1}(T \oplus V) = 0||Z \oplus X.$$

For any previous query $(K, 1||Z, V)$, the number of possible target values for $E_K^{-1}(T \oplus V)$ is at most q_h , since $\mathcal{A}$ makes at most q_h hash queries. As $\mathcal{A}$ makes at most q_p ideal cipher queries, the success probability in this case is at most $2q_p q_h / 2^n$.

If the hash query producing X occurs after the inverse query, then $\mathcal{A}$ must find a preimage in a set of size at most q_p . In that case, the success probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; q_p)$.

Case 2: The ideal cipher query involving $(K, 1||Z, V)$ occurs after the query involving $(K, 0||N, U)$.

By a symmetric argument, the success probability in this case is bounded by

$$\frac{2q_p^2}{2^n} + \frac{2q_p q_h}{2^n} + \text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; q_p).$$

Combining both cases, we have the theorem.

A matching attack for CDY_k is presented in Appendix D.2, establishing the tightness for both CDY_k and CDY .

8 Committing Security of MACs from DbHtS Paradigm

In this section, we analyze the committing security of four beyond-birthday-bound secure MACs within DbHtS paradigm: 3kf9, PMAC_Plus, SUM-ECBC, and LightMAC_Plus.

8.1 3kf9

Zhang et al. [44] proposed a CBC-based MAC, 3kf9, obtained by combining f9 [1] and EMAC [34]. 3kf9 offers BBB PRF-security when its underlying n -bit blockcipher is pseudorandom with three independent keys.

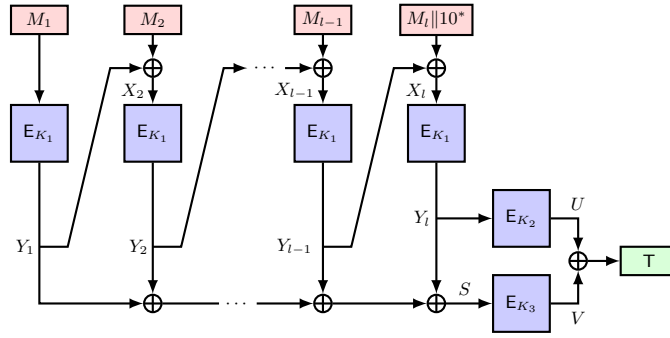


Fig. 22: 3kf9

8.1.1 CMT, CMT_k and RBT_k Security of 3kf9 We present a key-robustness attack on the 3kf9 construction. The adversary selects a key K_1 and two distinct keys K_2 and K_3 . It sets $M_1 := E_{K_1}^{-1}(0^n)$ and picks an arbitrary $M_2 \in \{0, 1\}^n$, forming $M = M_1 || M_2$. The adversary then outputs (K_1, K_2, K_2, M) and (K_1, K_3, K_3, M) . The complete attack is shown in Figure 23.

- 1 : Select a key $K_1 \in \{0, 1\}^\kappa$;
- 2 : Select two keys $K_2, K_3 \in \{0, 1\}^\kappa$ such that $K_2 \neq K_3$;
- 3 : Compute $M_1 = E_{K_1}^{-1}(0^n)$;
- 4 : Select any $M_2 \in \{0, 1\}^n$ and set $M = M_1 || M_2$;
- 5 : Output the tuples (K_1, K_2, K_2, M) and (K_1, K_3, K_3, M) ;

Fig. 23: RBT_k adversary against 3kf9

Analysis of the Attack: For any (K, K', K', M) , where $M = M_1 || M_2$ and $E_K(M_1) = 0^n$, we have $3kf9(K, K', K', M) = 0^n$. Thus, the algorithm presented in Figure 23 leads to a RBT_k attack on 3kf9 with a success probability of 1.

Therefore, the construction 3kf9 is not RBT_k secure and by Lemma 1 we can also conclude that 3kf9 is not CMT_k and CMT secure.

8.1.2 CDY Security of 3kf9 Here, we describe a CDY attack against the 3kf9 construction for any context selector S . The idea is to find context tuple (K_1, K_2, K_3, M) that produce the desired tag T (S 's output). The complete attack is provided in Figure 24.

- 1: Select any K_1, K_2 and $K_3 \in \{0, 1\}^\kappa$;
- 2: Select any $U \in \{0, 1\}^n$ and set $V = T \oplus U$;
- 3: Compute $Y_2 = E_{K_2}^{-1}(U)$, $S = E_{K_3}^{-1}(V)$ and set $Y_1 = S \oplus Y_2$;
- 4: Compute $M_1 = E_{K_1}^{-1}(Y_1)$, $X_2 = E_{K_1}^{-1}(Y_2)$ and Set $M_2 = X_2 \oplus Y_1$;
- 5: Output the tuple (K_1, K_2, K_3, M) , where $M = M_1 \| M_2$;

Fig. 24: CDY adversary against 3kf9

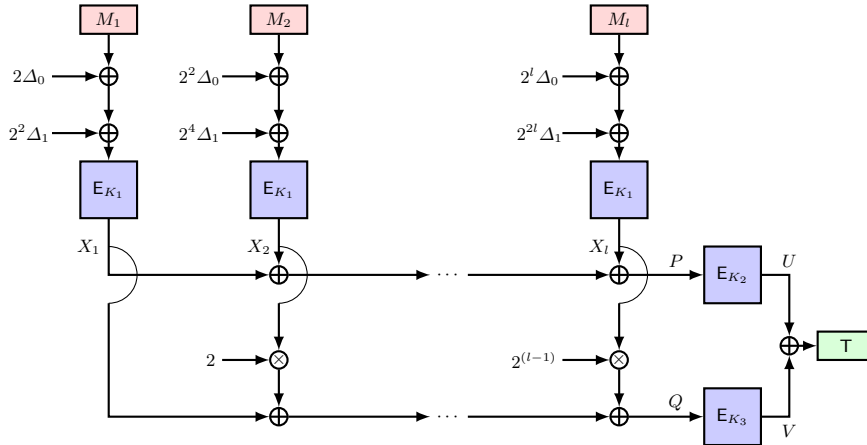
Analysis of the Attack: For the output tuple (K_1, K_2, K_3, M) , the tag is computed as:

$$\begin{aligned}
 & 3kf9(K_1, K_2, K_3, M) \\
 &= E_{K_2}(E_{K_1}(M_2 \oplus E_{K_1}(M_1))) \oplus E_{K_3}(E_{K_1}(M_2 \oplus E_{K_1}(M_1)) \oplus E_{K_1}(M_1)), \\
 &= E_{K_2}(E_{K_1}(M_2 \oplus Y_1)) \oplus E_{K_3}(E_{K_1}(M_2 \oplus Y_1) \oplus Y_1), \\
 &= E_{K_2}(E_{K_1}(X_2)) \oplus E_{K_3}(E_{K_1}(X_2) \oplus Y_1), \\
 &= E_{K_2}(Y_2) \oplus E_{K_3}(Y_2 \oplus Y_1) = E_{K_2}(Y_2) \oplus E_{K_3}(S) = U \oplus V = T.
 \end{aligned}$$

Thus, from the above calculation, it is clear that the tuple (K_1, K_2, K_3, M) outputs the desired tag T .

8.2 PMAC_Plus

Yasuda [43] proposed a variant of the PMAC construction, called PMAC_Plus, which achieves security against $O(2^{2n/3})$ queries while retaining rate-1 efficiency.

Fig. 25: PMAC_Plus. $\Delta_0 = E_{K_1}(0)$ and $\Delta_1 = E_{K_1}(1)$

8.2.1 CMT, CMT_k and RBT_k security of PMAC_Plus Now, we present a key-robustness attack against the PMAC_Plus construction. The adversary selects a key K_1 , and two distinct keys K_2 and K_3 . Next, the adversary computes, $\Delta_0 = E_{K_1}(0^n)$ and $\Delta_1 = E_{K_1}(1^n)$. For simplicity of our explanation, we consider $\alpha_1 = ((2 \otimes \Delta_0) \oplus (2^2 \otimes \Delta_1))$ and $\alpha_2 = ((2^2 \otimes \Delta_0) \oplus (2^4 \otimes \Delta_1))$. Next, the adversary fixes $M_2 := (\alpha_2 \oplus E_{K_1}^{-1}(0^n))$ and considers an arbitrary M_1 , then sets $M = M_1 \| M_2$. Finally, it outputs the tuples (K_1, K_2, K_2, M) and (K_1, K_3, K_3, M) . The complete attack is provided in Figure 26.

```

1 : Select a key  $K_1 \in \{0, 1\}^\kappa$ ;
2 : Select two keys  $K_2, K_3 \in \{0, 1\}^\kappa$  such that  $K_2 \neq K_3$ ;
3 : Compute  $M_2 = \alpha_2 \oplus E_{K_1}^{-1}(0^n)$ , where  $\alpha_2 = ((2^2 \otimes \Delta_0) \oplus (2^4 \otimes \Delta_1))$ ;
4 : Choose any  $M_1 \in \{0, 1\}^n$  and set  $M = M_1 \| M_2$ ;
5 : Output the tuples  $(K_1, K_2, K_2, M)$  and  $(K_1, K_3, K_3, M)$ ;

```

Fig. 26: RBT_k adversary against PMAC_Plus

Analysis of the Attack: For any (K, K', K', M) , where $M = M_1 \| M_2$ and $E_K(M_2 \oplus \alpha_2) = 0^n$, we have $\text{PMAC_Plus}(K, K', K', M) = 0^n$. Thus, the algorithm presented in Figure 26 leads to a RBT_k attack on PMAC_Plus with a success probability of 1.

Therefore, the construction PMAC_Plus is not RBT_k secure and using Lemma 1 we get that it is also not CMT_k and CMT secure.

8.2.2 CDY Security of PMAC_Plus We describe an CDY attack against the PMAC_Plus construction for any context selector S . The idea is to find a context tuple (K_1, K_2, K_3, M) that produce the desired tag T (S 's output). The complete attack is provided in Figure 27.

```

1 : Choose arbitrary  $K_1, K_2$  and  $K_3 \in \{0, 1\}^\kappa$ ;
2 : Select any  $U \in \{0, 1\}^n$  and compute  $V = T \oplus U$ ;
3 : Compute  $P = E_{K_2}^{-1}(U)$  and  $Q = E_{K_3}^{-1}(V)$ ;
4 : Find  $X_1, X_2$  such that  $X_1 \oplus X_2 = P$  &  $X_1 \oplus (2 \otimes X_2) = Q$ ;
5 : Compute  $M_1 = E_{K_1}^{-1}(X_1) \oplus \alpha_1$ , where,  $\alpha_1 = ((2 \otimes \Delta_0) \oplus (2^2 \otimes \Delta_1))$ ;
6 : Compute  $M_2 = E_{K_1}^{-1}(X_2) \oplus \alpha_2$ , where,  $\alpha_2 = ((2^2 \otimes \Delta_0) \oplus (2^4 \otimes \Delta_1))$ ;
7 : Output the tuple  $(K_1, K_2, K_3, M)$ , where  $M = M_1 \| M_2$ ;

```

Fig. 27: CDY adversary against PMAC_Plus

Analysis of the attack: Here, the main step is to solve the system of equations:

$$\begin{aligned} X_1 \oplus X_2 &= P, \\ X_1 \oplus (2 \otimes X_2) &= Q. \end{aligned}$$

This system of equations has a unique solution as the corresponding coefficient matrix has rank 2. Thus, we have shown a valid CDY attack against the PMAC_Plus construction with success probability 1.

Remark 1. This attack relies on finding a preimage for PHash: for any $P \| Q$, a (K_1, M) pair can be found with a constant number of ideal cipher queries. Using the same approach, a constant-query CDY attack on 4F-PHash-DbHtF is possible (see Appendix E.2.2). Similarly, we present a CMT_k attack on 4F-PHash-DbHtF in Appendix E.2.1.

8.3 SUM-ECBC

Yasuda [42] proposed a deterministic, stateless, block-cipher-based MAC, SUM-ECBC, and proved that it achieves PRF security against $O(2^{2n/3})$ queries, assuming the underlying n -bit block cipher is a secure pseudorandom permutation.

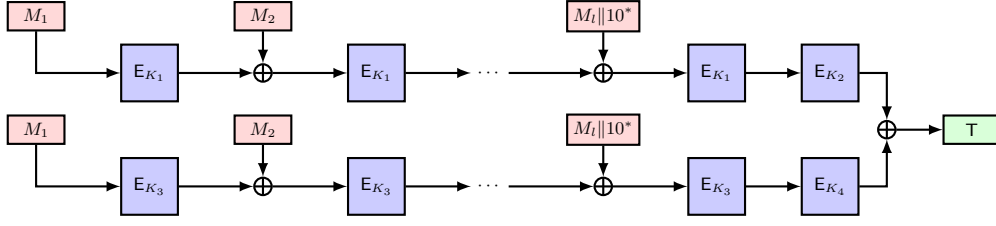


Fig. 28: SUM-ECBC

8.3.1 CMT, CMT_k and RBT_k security of SUM-ECBC Here, we present a key-robustness attack against the SUM-ECBC construction. The adversary selects a key K_1 , and two distinct keys K_2 and K'_2 . Next, the adversary considers an arbitrary M_1 and M_2 , then sets $M = M_1 || M_2$. Finally, it outputs the tuples (K_1, K_2, K_1, K_2, M) and $(K_1, K'_2, K_1, K'_2, M)$. The complete attack is provided in Figure 29.

- 1 : Select three keys $K_1, K_2, K'_2 \in \{0, 1\}^\kappa$ such that $K_2 \neq K'_2$;
- 2 : Choose arbitrary M_1 and $M_2 \in \{0, 1\}^n$ and set $M = M_1 || M_2$;
- 3 : Output the tuples (K_1, K_2, K_1, K_2, M) and $(K_1, K'_2, K_1, K'_2, M)$;

Fig. 29: RBT_k adversary against SUM-ECBC

Analysis of the Attack: For any (K, K', K, K', M) , we have $\text{SUM-ECBC}(K, K', K, K', M) = 0^n$, where $M = M_1 || M_2$. Thus, the algorithm presented in Figure 29 leads to an RBT_k attack on SUM-ECBC with a success probability of 1.

Therefore, the construction SUM-ECBC is not RBT_k secure and using Lemma 1 we get that it is also not CMT_k and CMT secure.

8.4 LightMAC_Plus

Naito [33] proposed a variant of the LightMAC construction, called LightMAC_Plus, which breaks the birthday bound by achieving $O(2^{2n/3})$ security.

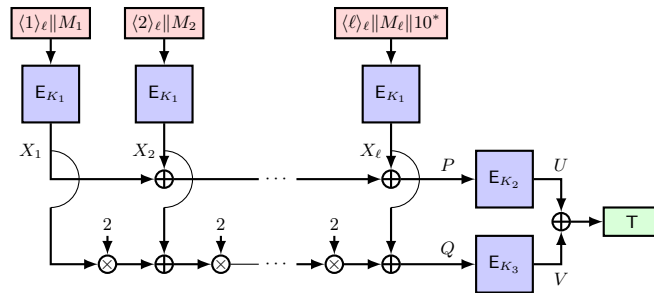


Fig. 30: LightMAC_Plus

8.4.1 CMT, CMT_k and RBT_k security of LightMAC_Plus We present a key-robustness attack against the LightMAC_Plus construction. Here, each block of message is of size $(n - \ell)$ -bits and the i -th block M_i is padded as $\langle i \rangle_\ell || M_i$, where $\langle i \rangle_\ell$ is ℓ -bit binary representation of integer i . The complete attack strategy is described in Figure 31.

```

1 : Select a key  $K_1 \in \{0, 1\}^\kappa$ ;
2 : Select two keys  $K_2, K_3 \in \{0, 1\}^\kappa$  such that  $K_2 \neq K_3$ ;
3 : Compute  $A = E_{K_1}^{-1}(0^n)$ ;
4 : Set  $M_1 = \text{lsb}_{n-\ell}(A)$ ;
5 : Choose arbitrary  $M_2 \in \{0, 1\}^{n-\ell}$  and set  $M = M_1 \| M_2$ ;
6 : Output the tuples  $(K_1, K_2, K_2, M)$  and  $(K_1, K_3, K_3, M)$ ;

```

Fig. 31: RBT_k adversary against LightMAC_Plus

Analysis and Success Probability of the attack: Whenever $\langle 1 \rangle_\ell \| M_1 = E_K^{-1}(0^n)$ holds, we have $\text{LightMAC_Plus}(K, K', K', M_1 \| M_2) = 0^n$ for any tuple $(K, K', K', M_1 \| M_2)$. Under the ideal-cipher assumption, the value $E_{K_1}^{-1}(0^n)$ is uniformly distributed over $\{0, 1\}^n$. So, we have

$$\Pr[E_{K_1}^{-1}(0^n) = \langle 1 \rangle_\ell \| *] = \frac{1}{2^\ell}.$$

Hence, the overall success probability of the adversary is $2^{-\ell}$.

8.4.2 CDY Security of LightMAC_Plus We describe a CDY attack against the LightMAC_Plus construction for any context selector S . The idea is to find a context tuple (K_1, K_2, K_3, M) that produces the desired tag $T(S$'s output). The complete attack is provided in Figure 32.

```

1 : Choose arbitrary  $K_1, K_2$  and  $K_3 \in \{0, 1\}^\kappa$ ;
2 : Select arbitrary  $U \in \{0, 1\}^n$  and compute  $V = T \oplus U$ ;
3 : Compute  $P = E_{K_2}^{-1}(U)$  and  $Q = E_{K_3}^{-1}(V)$ ;
4 : Solve for the system of equations:  $X_1 \oplus X_2 = P$  &  $(2 \otimes X_1) \oplus X_2 = Q$ ;
5 : Compute  $A = E_{K_1}^{-1}(X_1)$  and  $B = E_{K_1}^{-1}(X_2)$ ;
6 : Set  $M_1 = \text{lsb}_{n-\ell}(A)$  and  $M_2 = \text{lsb}_{n-\ell}(B)$ ;
7 : Set  $M = M_1 \| M_2$ ;
8 : Output the tuple  $(K_1, K_2, K_3, M)$ ;

```

Fig. 32: CDY adversary against LightMAC_Plus

Analysis and Success Probability of the attack: The adversary will be successful if and only if $\langle 1 \rangle_\ell \| * = E_{K_1}^{-1}(X_1)$ and $\langle 2 \rangle_\ell \| * = E_{K_1}^{-1}(X_2)$

For the above attack, we require that $\langle 1 \rangle_\ell \| M_1 = E_{K_1}^{-1}(X_1)$ and $\langle 2 \rangle_\ell \| * = E_{K_1}^{-1}(X_2)$. Under the ideal-cipher assumption, the values $E_{K_1}^{-1}(X_1)$ and $E_{K_1}^{-1}(X_2)$ are uniformly distributed over $\{0, 1\}^n$. So,

$$\Pr[\langle 1 \rangle_\ell \| * = E_{K_1}^{-1}(X_1) \wedge \langle 2 \rangle_\ell \| * = E_{K_1}^{-1}(X_2)] \leq \frac{2}{2^{2\ell}}$$

So, the adversary's success probability is upper bounded by $2^{-2\ell}$.

Moreover, the system of equations

$$X_1 \oplus X_2 = P, \quad (2 \otimes X_1) \oplus X_2 = Q.$$

has a unique solution as the corresponding coefficient matrix has rank 2. Thus, we have shown a valid CDY attack against the LightMAC_Plus construction with success probability $2^{-2\ell}$.

Remark 2. Similar to PMAC_Plus , the CDY attack on LightHash exploits preimage finding. Using the same approach as 4F-PHash-DbHtF , constant query attacks for both CMT_k and CDY can be mounted with success probability $1/2^{2\ell}$ on $4\text{F-LightHash-DbHtF}$.

9 Conclusion

Among the classified BBB-secure schemes and nEHtM, only nEHtM achieves non-trivial CMT_k and CMT security, both up to $2^{n/4}$, with no matching tight attack; all others fail to satisfy these notions. Several schemes achieve tight $2^{n/2}$ security for both CDY_k and CDY, while others are limited to $2^{n/3}$ or remain unresolved. Tightness is established for most constructions via matching attacks, with a few cases still open. In nearly all schemes, CDY_k and CDY coincide, with only minor separations observed. RBT_k security ranges from absent to $2^{n/4}$, depending on the construction.

For constructions following the DbHtS paradigm, we provide constant-query attacks on RBT_k for three constructions (also applying to CMT_k and CMT) and on CDY for two schemes; one allows RBT_k and CDY attacks with success probability $2^{-\ell}$ and $2^{-2\ell}$ respectively, practical for short counters. These results demonstrate that subtle design differences among BBB MACs lead to markedly different committing security profiles. A promising direction for future work is exploring the impact of key derivation, as many attacks on CMT_k and CMT fail when keys are derived from a master key, though not universally, as evidenced by our CMT_k attack on PMAC_Plus (Appendix E.1). Another avenue is investigating whether Davies-Meyer style transformations, which enhance committing security in prior work such as [6], can similarly strengthen these BBB constructions.

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A Auxiliary Analysis of EDM-based Constructions

A.1 Analysis of the CMT_k Attack on $F_{B_2}^{\text{EDM}}$

For the first output tuple (K, K, K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_2}^{\text{EDM}}(K, K, K_h, N, M) &= \mathbf{E}_K(N \oplus \mathbf{E}_K(N) \oplus \mathbf{H}(K_h, M)), \\ &= \mathbf{E}_K(N \oplus X \oplus X), \text{ (as } X = \mathbf{H}(K_h, M), N = \mathbf{E}_K^{-1}(X)) \\ &= \mathbf{E}_K(N) = X. \end{aligned}$$

For the second output tuple (K', K', K_h, N', M) , the tag is computed as:

$$\begin{aligned} F_{B_2}^{\text{EDM}}(K', K', K_h, N', M) &= \mathbf{E}_{K'}(N' \oplus \mathbf{E}_{K'}(N') \oplus \mathbf{H}(K_h, M)), \\ &= \mathbf{E}_{K'}(N' \oplus X \oplus X), \text{ (as } X = \mathbf{H}(K_h, M), N' = \mathbf{E}_{K'}^{-1}(X)) \\ &= \mathbf{E}_{K'}(N') = X. \end{aligned}$$

From the above two equations, it is clear that both tuples produce the same tag. Thus, the algorithm presented in Figure 6 leads to a CMT_k attack on $F_{B_2}^{\text{EDM}}$ with a success probability of 1.

A.2 Matching attack for RBT_k on $F_{B_2}^{\text{EDM}}$

We now demonstrate the tightness of the RBT_k bound for $F_{B_2}^{\text{EDM}}$ as established in the preceding theorem. The attack proceeds as follows:

- Select $2^{n/4}$ distinct values for K_1 . Fix a nonce N and a message M . For each $i \in [1, 2^{n/4}]$, compute $U_i = \mathbf{E}_{K_1^i}(N)$.
- Independently, select $2^{n/4}$ distinct values for K_h , and for each $i \in [1, 2^{n/4}]$, compute $X_i = \mathbf{H}(K_h^i, M)$.
- Find indices $a, b, c, d \in [1, 2^{n/4}]$ such that $U_a \oplus X_b = U_c \oplus X_d$.
- Fix a key K_2 arbitrarily.
- Output the tuple $((K_1^a, K_2, K_h^b), (K_1^c, K_2, K_h^d), N, M)$.

Since each U_i and X_i is sampled uniformly at random and independently, the success probability of the attack is at least $1 - e^{-1}$.

A.3 Matching CDY_k Attack on $F_{B_2}^{\text{EDM}}$

Here, we present a matching attack on CDY_k, which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_2}^{\text{EDM}}$. Consider any challenge tuple (N, M, T) . The adversary proceeds as follows:

- Sample distinct $K_1^i \leftarrow \{0, 1\}^\kappa$, $\forall i \in [1, 2^{n/3}]$, and compute $U_i = \mathbf{E}_{K_1^i}(N)$.
- Sample distinct $K_2^i \leftarrow \{0, 1\}^\kappa$, $\forall i \in [1, 2^{n/3}]$, and compute $V_i = \mathbf{E}_{K_2^i}^{-1}(T)$.
- Sample distinct $K_h^i \leftarrow \{0, 1\}^\kappa$, $\forall i \in [1, 2^{n/3}]$, and compute $X_i = \mathbf{H}(K_h^i, M)$.
- Search for indices $a, b, c \in [1, 2^{n/3}]$ such that $U_a \oplus V_b = X_c$.
- Output the key triple (K_1^a, K_2^b, K_h^c) .

Since all values U_i , V_i , and X_i are independently and uniformly distributed (with replacement), the probability that such a triple (a, b, c) exists is at least $1 - e^{-1}$. Therefore, the adversary wins with probability at least $1 - e^{-1}$.

A.4 Analysis of the CMT_k Attack on $F_{B_3}^{\text{EDM}}$

For the first tuple (K, K, K_h, N, M) , the computed tag is:

$$\begin{aligned} F_{B_3}^{\text{EDM}}(K, K, K_h, N, M) &= \mathbf{E}_K(N \oplus \mathbf{E}_K(N) \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M) \\ &= \mathbf{E}_K(N \oplus X \oplus X) \oplus X \text{ (as } X = \mathbf{H}(K_h, M), N = \mathbf{E}_K^{-1}(X)) \\ &= \mathbf{E}_K(N) \oplus X = X \oplus X = 0. \end{aligned}$$

Similarly, for the second tuple (K', K', K_h, N', M) , the tag is:

$$\begin{aligned} F_{B_3}^{\text{EDM}}(K', K', K_h, N', M) &= \mathbf{E}_{K'}(N' \oplus \mathbf{E}_{K'}(N') \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M) \\ &= \mathbf{E}_{K'}(N' \oplus X \oplus X) \oplus X \text{ (as } N' = \mathbf{E}_{K'}^{-1}(X)) \\ &= \mathbf{E}_{K'}(N') \oplus X = X \oplus X = 0. \end{aligned}$$

Hence, both generated tuples yield the same tag. Thus, the adversary described in Figure 8 successfully performs a CMT_k attack on $F_{B_3}^{\text{EDM}}$ with success probability 1.

A.5 Proof of Theorem 5

Let $\mathcal{A}$ output $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that (i) $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$ and (ii) $F_{B_3}^{\text{EDM}}(K_1, K_2, K_h, N, M) = F_{B_3}^{\text{EDM}}(K'_1, K'_2, K'_h, N, M)$.

Define

$$\mathbf{T} := \mathbf{E}_{K_2}(\mathbf{E}_{K_1}(N) \oplus N \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M),$$

$$\mathbf{T}' := \mathbf{E}_{K'_2}(\mathbf{E}_{K'_1}(N) \oplus N \oplus \mathbf{H}(K'_h, M)) \oplus \mathbf{H}(K'_h, M).$$

The adversary succeeds if and only if $\mathbf{T} = \mathbf{T}'$. We distinguish two cases.

Case 1: $K_h = K'_h$. Then success implies $\mathbf{E}_{K_2}(V) = \mathbf{E}_{K'_2}(V')$, where

$$V = \mathbf{E}_{K_1}(N) \oplus N \oplus \mathbf{H}(K_h, M), \quad V' = \mathbf{E}_{K'_1}(N) \oplus N \oplus \mathbf{H}(K_h, M).$$

1a: If $K_2 = K'_2$, then $V = V'$ and success requires $\mathbf{E}_{K_1}(N) = \mathbf{E}_{K'_1}(N)$ with $K_1 \neq K'_1$. By ideal-cipher randomness, this occurs with probability at most $2q_p^2/2^n$.

1b: If $K_2 \neq K'_2$, let the query involving K'_2 occur last.

1b1: If it is forward, the success probability is at most $2q_p^2/2^n$.

1b2: If it is inverse, then

$$\mathbf{E}_{K'_2}^{-1}(W') = \mathbf{E}_{K_1}(N) \oplus N \oplus \mathbf{H}(K_h, M).$$

If an ideal-cipher query is last, the probability is at most $2q_p^2q_h/2^n$. If the hash query is last, success requires

$$\mathbf{H}(K_h, M) \in \{\mathbf{E}_{K'_2}^{-1}(W') \oplus \mathbf{E}_{K_1}(N) \oplus N\},$$

whose size is at most q_p^2 ; hence the probability is bounded by $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2)$.

Case 2: $K_h \neq K'_h$.

2a: If $\mathbf{H}(K_h, M) = \mathbf{H}(K'_h, M)$, this yields a hash collision; the probability is bounded by $\mathbf{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$.

2b: If $\mathbf{H}(K_h, M) \neq \mathbf{H}(K'_h, M)$, then

$$\mathbf{E}_{K_2}(V) \oplus \mathbf{H}(K_h, M) = \mathbf{E}_{K'_2}(V') \oplus \mathbf{H}(K'_h, M).$$

2b1: If a forward ideal-cipher query is last among the relevant queries, the probability is at most $2q_p^2q_h^2/2^n$.

2b2: If an inverse ideal-cipher query is the final query, then by the same argument as in Case 1b2, the adversary's success probability is bounded by $2q_p^2q_h/2^n + \mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2)$.

2b3: If the hash query is last, success requires

$$\mathbf{H}(K'_h, M) \in \{\mathbf{E}_{K_2}(V) \oplus \mathbf{H}(K_h, M) \oplus \mathbf{E}_{K'_2}(V')\},$$

a set of size at most $q_p^2q_h$, yielding bound $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2q_h)$.

Collecting the bounds from all cases gives the stated result.

A.6 RBT_k Attack on $F_{B_3}^{\text{EDM}}$

We now demonstrate an RBT_k attack with $O(2^{n/2})$ queries for the $F_{B_3}^{\text{EDM}}$ construction. The attack proceeds as follows:

- Fix a key K_h and a message M . Compute $X = \mathbf{H}(K_h, M)$.
- Fix a key K_1 and a nonce N . Compute $U = \mathbf{E}_{K_1}(N)$ and $V = U \oplus N \oplus X$.
- Select $2^{n/2}$ distinct values for K_2 . For each $i \in [1, 2^{n/2}]$, compute $\mathbf{T}_i = \mathbf{E}_{K_2^i}(V) \oplus X$.
- Find indices $a, b \in [1, 2^{n/2}]$ such that $\mathbf{T}_a = \mathbf{T}_b$.
- Output the tuple $((K_1, K_2^a, K_h), (K_1, K_2^b, K_h), N, M)$.

Since each $\mathbf{T}_i$ is sampled independently and uniformly at random, the success probability of the attack is at least $1 - e^{-1}$.

A.7 Proof of Theorem 6

Let T be the challenge tag and $\mathcal{A}$ outputs (K_1, K_2, K_h, N, M) such that $F_{B_3}^{\text{EDM}}(K_1, K_2, K_h, N, M) = T$. If $T = 0^n$, the adversary can always construct a valid context: choose arbitrary K_h, M and compute $X = H(K_h, M)$; choose arbitrary K and set $N = E_K^{-1}(X)$. Then

$$F_{B_3}^{\text{EDM}}(K, K, K_h, N, M) = E_K(E_K(N) \oplus N \oplus X) \oplus X = E_K(N) \oplus X = 0^n.$$

Thus the success probability in this case is bounded by ϵ_0 .

Hence assume $T \neq 0^n$. If $(K_1, N) = (K_2, V)$, then necessarily $T = 0^n$; thus we may assume $(K_1, N) \neq (K_2, V)$. Let $V = E_{K_1}(N) \oplus N \oplus H(K_h, M)$. $\mathcal{A}$'s Success requires

$$E_{K_2}(V) \oplus H(K_h, M) = T.$$

Case 1: The hash query on (K_h, M) occurs after the ideal-cipher query on (K_2, V, W) . Then $\mathcal{A}$'s success requires

$$H(K_h, M) \in \{W \oplus T\},$$

where $W = E_{K_2}(V)$ and there are at most q_p possible values of W . Hence, the success probability is bounded by $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; q_p)$.

Case 2: A forward ideal-cipher query on (K_2, V) occurs after the hash query. By ideal-cipher randomness over at most $q_p q_h$ combinations, the probability that

$$E_{K_2}(V) \oplus H(K_h, M) = T$$

holds is at most $2q_p q_h / 2^n$.

Case 3: An inverse ideal-cipher query on (K_2, W) occurs after the hash query. Success requires

$$E_{K_2}^{-1}(W) = E_{K_1}(N) \oplus N \oplus T \oplus W,$$

which, by ideal-cipher randomness and at most q_p such queries, occurs with probability at most $2q_p^2 / 2^n$. Combining all cases yields the stated bound.

A.8 Matching CDY_k Attack on $F_{B_3}^{\text{EDM}}$

Now, we present a matching attack on CDY_k, which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_3}^{\text{EDM}}$. Consider any challenge tuple (N, M, T) . The adversary proceeds as follows:

- Choose $2^{n/2}$ distinct $K_1^i \leftarrow \{0, 1\}^\kappa$, $\forall i \in [1, 2^{n/2}]$, and compute $U_i = E_{K_1^i}(N)$.
- Choose $K_h \leftarrow \{0, 1\}^\kappa$ and compute $X = H(K_h, M)$.
- Choose $2^{n/2}$ distinct $K_2^i \leftarrow \{0, 1\}^\kappa$, $\forall i \in [1, 2^{n/2}]$, and compute $V_i = E_{K_2^i}^{-1}(T \oplus X)$.
- Find $a, b \in [1, 2^{n/2}]$ such that $V_b = U_a \oplus N \oplus X$.
- Output the key triple (K_1^a, K_h, K_2^b) .

Since all values U_i, V_i are independently and uniformly distributed (with replacement), the probability that such a tuple (a, b) exists is at least $1 - e^{-1}$. Therefore, the adversary wins with probability at least $1 - e^{-1}$.

A.9 Analysis of the CMT_k Attack on $F_{B_4}^{\text{EDM}}$

For the first output tuple (K, K, K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_4}^{\text{EDM}}(K, K, K_h, N, M) &= E_K(N \oplus E_K(N \oplus H(K_h, M))) \\ &= E_K(N \oplus E_K(E_K^{-1}(X))) \\ &= E_K(N \oplus X) = E_K(E_K^{-1}(X)) = X. \end{aligned}$$

where we use $X = H(K_h, M)$ and $N = E_K^{-1}(X) \oplus X$.

For the second output tuple (K', K', K_h, N', M) , the tag is computed as:

$$\begin{aligned} F_{B_4}^{\text{EDM}}(K', K', K_h, N', M) &= E_{K'}(N' \oplus E_{K'}(N' \oplus H(K_h, M))) \\ &= E_{K'}(N' \oplus E_{K'}(E_{K'}^{-1}(X))) \\ &= E_{K'}(N' \oplus X) = E_{K'}(E_{K'}^{-1}(X)) = X. \end{aligned}$$

where we use $X = H(K_h, M)$ and $N' = E_{K'}^{-1}(X) \oplus X$.

From the above computations, it is clear that both tuples yield the same tag X . Thus, the algorithm described in Figure 10 constitutes a CMT_k attack on $F_{B_4}^{\text{EDM}}$ with success probability 1.

A.10 Matching attack for RBT_k on $F_{B_4}^{\text{EDM}}$

We now demonstrate the tightness of the above bound via the following attack:

- Fix any message M .
- Choose $2^{n/4}$ distinct keys K_h^i and compute $X_i = H(K_h^i, M)$, $\forall i \in [1, 2^{n/4}]$.
- Fix any U .
- Choose $2^{n/4}$ distinct keys K_1^i and compute $Z_i = E_{K_1^i}^{-1}(U)$, $\forall i \in [1, 2^{n/4}]$.
- Find $a, b, c, d \in [1, 2^{n/4}]$ such that $Z_a \oplus X_b = Z_c \oplus X_d$.
- Fix any K_2 . Compute $N = Z_a \oplus X_b$.
- Output $((K_1^a, K_2, K_h^b), (K_1^c, K_2, K_h^d), N, M)$.

Since each Z_i is independently and uniformly random, the collision probability is at least $1 - e^{-1}$.

A.11 Matching Attack for CDY on $F_{B_4}^{\text{EDM}}$

We demonstrate that the bound given in the previous theorem for $F_{B_4}^{\text{EDM}}$ is tight by describing a concrete attack strategy:

- Choose distinct $K_1^i \leftarrow \{0, 1\}^\kappa$ for all $i \in [1, 2^{n/3}]$, and compute $U_i = E_{K_1^i}(Z)$ for a fixed $Z \in \{0, 1\}^n$.
- Choose distinct $K_h^i \leftarrow \{0, 1\}^{\kappa_h}$ for all $i \in [1, 2^{n/3}]$, and compute $X_i = H(K_h^i, M)$ for a fixed message M .
- Choose distinct $K_2^i \leftarrow \{0, 1\}^\kappa$ for all $i \in [1, 2^{n/3}]$, and compute $V_i = E_{K_2^i}^{-1}(T)$.
- Find $a, b, c \in [1, 2^{n/3}]$ such that:

$$V_c = U_a \oplus X_b \oplus Z.$$

- Output the context $(K_1^a, K_2^c, K_h^b, N, M)$ where $N = X_b \oplus Z$.

Since all values U_i , X_i , and V_i are independently and uniformly distributed (with replacement), the probability that such a triple (a, b, c) exists is at least $1 - e^{-1}$. Hence, the adversary succeeds with probability at least $1 - e^{-1}$.

A.12 Matching Attack for CDY_k on $F_{B_4}^{\text{EDM}}$

Here, we demonstrate that the CDY_k bound in the previous theorem is tight. The attack proceeds as follows.

- Choose any $K_2 \leftarrow \{0, 1\}^\kappa$, and compute $V = E_{K_2}^{-1}(T)$.
- Choose distinct $K_h^i \leftarrow \{0, 1\}^{\kappa_h}$ for all $i \in [1, 2^{n/2}]$, and compute $X_i = H(K_h^i, M)$.
- Choose distinct $K_1^i \leftarrow \{0, 1\}^\kappa$ for all $i \in [1, 2^{n/2}]$, and compute $Z_i = E_{K_1^i}^{-1}(V \oplus N)$.
- Find $a, b \in [1, 2^{n/2}]$ such that $Z_b = X_a \oplus N$.
- Output the key (K_1^a, K_h^b, K_2) .

Since all values Z_i , X_i are independently and uniformly distributed (with replacement), the probability that such a tuple (a, b, c) exists is at least $1 - e^{-1}$. Therefore, the adversary wins with probability at least $1 - e^{-1}$.

A.13 Analysis of the CMT_k Attack on $F_{B_5}^{\text{EDM}}$

For the first output tuple (K, K, K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_5}^{\text{EDM}}(K, K, K_h, N, M) &= E_K(N \oplus E_K(N \oplus H(K_h, M))) \oplus H(K_h, M), \\ &= E_K(N \oplus E_K(E_K^{-1}(X))) \oplus X, \\ &= E_K(N \oplus X) \oplus X = E_K(E_K^{-1}(X)) \oplus X = 0. \end{aligned}$$

where we use the values $X = H(K_h, M)$ and $N = E_K^{-1}(X) \oplus X$.

For the second output tuple (K', K', K_h, N', M) , the tag is computed as:

$$\begin{aligned} F_{B_5}^{\text{EDM}}(K', K', K_h, N', M) &= E_{K'}(N' \oplus E_{K'}(N' \oplus H(K_h, M))) \oplus H(K_h, M), \\ &= E_{K'}(N' \oplus E_{K'}(E_{K'}^{-1}(X))) \oplus X, \\ &= E_{K'}(N' \oplus X) \oplus X = E_{K'}(E_{K'}^{-1}(X)) \oplus X = 0. \end{aligned}$$

where we use the values $X = H(K_h, M)$ and $N' = E_{K'}^{-1}(X) \oplus X$.

From the above two equations, it is clear that both tuples produce the same tag. Thus, the algorithm described in Figure 12 results in a CMT_k attack on $F_{B_5}^{\text{EDM}}$ with a success probability of 1.

A.14 Proof of Theorem 10

Let $\mathcal{A}$ output $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that

1. $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$, and
2. $F_{B_5}^{\text{EDM}}(K_1, K_2, K_h, N, M) = F_{B_5}^{\text{EDM}}(K'_1, K'_2, K'_h, N, M)$.

Define

$$\begin{aligned} \mathsf{T} &:= \mathsf{E}_{K_2}(\mathsf{E}_{K_1}(N \oplus \mathsf{H}(K_h, M)) \oplus N) \oplus \mathsf{H}(K_h, M), \\ \mathsf{T}' &:= \mathsf{E}_{K'_2}(\mathsf{E}_{K'_1}(N \oplus \mathsf{H}(K'_h, M)) \oplus N) \oplus \mathsf{H}(K'_h, M). \end{aligned}$$

The adversary succeeds only if $\mathsf{T} = \mathsf{T}'$. We analyze this event by distinguishing three cases.

Case 1: $K_h = K'_h$. In this case, $\mathsf{T} = \mathsf{T}'$ implies

$$\mathsf{E}_{K_2}(\mathsf{E}_{K_1}(N \oplus \mathsf{H}(K_h, M)) \oplus N) = \mathsf{E}_{K'_2}(\mathsf{E}_{K'_1}(N \oplus \mathsf{H}(K_h, M)) \oplus N),$$

which is exactly a key-robustness collision for $F_{B_4}^{\text{EDM}}$ under the common hash key K_h . Hence, in this case, the success probability is bounded by $\text{Adv}_{F_{B_4}^{\text{EDM}}}^{\text{RBT}_k}(\mathcal{A})$.

Case 2: $K_h \neq K'_h$ and $\mathsf{H}(K_h, M) = \mathsf{H}(K'_h, M)$. This is a hash-collision event. Therefore, the adversary's success probability in this case is bounded by $\text{Adv}_{\mathsf{H}}^{\text{CR}}(\mathcal{A}_{cr})$.

Case 3: $K_h \neq K'_h$ and $\mathsf{H}(K_h, M) \neq \mathsf{H}(K'_h, M)$. Let

$$V := \mathsf{E}_{K_1}(N \oplus \mathsf{H}(K_h, M)) \oplus N, \quad V' := \mathsf{E}_{K'_1}(N \oplus \mathsf{H}(K'_h, M)) \oplus N.$$

Then the adversary must satisfy

$$\mathsf{E}_{K_2}(V) \oplus \mathsf{H}(K_h, M) = \mathsf{E}_{K'_2}(V') \oplus \mathsf{H}(K'_h, M).$$

We distinguish subcases according to the last among the four relevant queries (two hash queries and two ideal-cipher queries).

- 3a. Suppose a hash query (w.l.o.g. (K'_h, M)) is the final query. Then $\mathcal{A}$ succeeds only if it finds a preimage of H for the target set $\{\mathsf{E}_{K_2}(V) \oplus \mathsf{H}(K_h, M) \oplus \mathsf{E}_{K'_2}(V')\}$. The cardinality of this set is at most $q_p^2 q_h$. Hence, the success probability is bounded by $\text{Adv}_{\mathsf{H}}^{\text{Pre}}(\mathcal{A}_{pre}; q_p^2 q_h)$.
- 3b. Suppose a forward ideal-cipher query is the final query. By the randomness of the ideal cipher, the probability that the required equality holds is at most $\frac{2q_p^2 q_h^2}{2^n}$.
- 3c. Suppose the final query is an inverse ideal-cipher query. This situation is analogous to the combined inverse-query subcases (Cases 2b–2d) in the RBT_k proof (theorem 7) of $F_{B_4}^{\text{EDM}}$. Therefore, the success probability in this case is bounded by $\frac{2q_p^2}{2^n} + \frac{2q_p^4}{2^n} + \frac{2q_p^3 q_h}{2^n}$.

Summing the bounds from the three cases yields the claimed result.

A.15 Matching attack for RBT_k on $F_{B_5}^{\text{EDM}}$

We now demonstrate the tightness of the above bound via the following attack:

- Fix a key K_h and message M . Compute $X = \mathsf{H}(K_h, M)$.
- Fix a key K_1 and nonce N . Compute $U = \mathsf{E}_{K_1}(N \oplus X)$ and $V = U \oplus N$.
- Select $2^{n/2}$ distinct values for K_2 . For each $i \in [1, 2^{n/2}]$, compute:

$$\mathsf{T}_i = \mathsf{E}_{K_2^i}(V) \oplus X.$$

- Find $a, b \in [1, 2^{n/2}]$ such that $\mathsf{T}_a = \mathsf{T}_b$.
- Output the tuple $((K_1, K_2^a, K_h), (K_1, K_2^b, K_h), N, M)$.

By the birthday bound, the probability that such a collision occurs is at least $1 - e^{-1}$.

A.16 Matching Attack for CDY_k on $F_{B_5}^{\text{EDM}}$

Now, we present a matching attack on CDY_k , which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_5}^{\text{EDM}}$. Consider any challenge tuple (N, M, T) . The adversary proceeds as follows:

- Choose $K_h \leftarrow \{0, 1\}^{\kappa_h}$ and compute $X = \text{H}(K_h, M)$.
- For all $i \in [1, 2^{n/2}]$, choose distinct $K_2^i \leftarrow \{0, 1\}^\kappa$ and compute

$$V_i = \text{E}_{K_2^i}^{-1}(\mathsf{T} \oplus X).$$

- Let $Z = N \oplus X$. For all $i \in [1, 2^{n/2}]$, choose distinct $K_1^i \leftarrow \{0, 1\}^\kappa$ and compute

$$U_i = \text{E}_{K_1^i}(Z).$$

- Find indices $a, b \in [1, 2^{n/2}]$ such that

$$U_a = V_b \oplus Z \oplus X = V_b \oplus N.$$

- Output the key triple (K_1^a, K_h, K_2^b) .

Since all values U_i and V_i are independently and uniformly distributed (with replacement), the probability that such a collision (a, b) exists is at least $1 - e^{-1}$. Therefore, the adversary succeeds with probability at least $1 - e^{-1}$.

B Auxiliary Analysis of EDMD-based Constructions**B.1 Tightness of the RBT_k bound of $F_{B_1}^{\text{EDMD}}$**

We now demonstrate that the bound in Theorem 12 is tight. The attack proceeds as follows:

- Fix a nonce N , a message M , and a key K_1 . Compute $U = \text{E}_{K_1}(N)$.
- Select $2^{n/4}$ distinct keys K_2^i and compute $W_i = \text{E}_{K_2^i}(U) \oplus U$, $\forall i \in [1, 2^{n/4}]$.
- Select $2^{n/4}$ distinct keys K_h^i , and compute $X_i = \text{H}(K_h^i, M)$, $\forall i \in [1, 2^{n/4}]$.
- Find $a, b, c, d \in [1, 2^{n/4}]$ such that $W_a \oplus X_b = W_c \oplus X_d$.
- Output $((K_1, K_2^a, K_h^b), (K_1, K_2^c, K_h^d), N, M)$.

Since each W_i is sampled independently and uniformly at random, the probability of finding such a collision is at least $1 - e^{-1}$.

B.2 Tightness of the CDY_k bound of $F_{B_1}^{\text{EDMD}}$

Now, we present a matching CDY_k attack on $F_{B_1}^{\text{EDMD}}$, which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_1}^{\text{EDMD}}$. Let (N, M, T) be the challenge tuple. The adversary proceeds as follows.

1. Fix any K_1 and compute $U = \text{E}_{K_1}(N)$.
2. Choose $2^{n/2}$ distinct keys K_2^i and compute $W_i = \text{E}_{K_2^i}(U) \oplus U$ for each $i \in [1, 2^{n/2}]$.
3. Choose $2^{n/2}$ distinct hash keys K_h^i and compute $X_i = \text{H}(K_h^i, M)$.
4. Search for indices $a, b \in [1, 2^{n/2}]$ such that

$$W_a \oplus X_b = \mathsf{T}.$$

5. Output the key tuple (K_1, K_2^a, K_h^b) .

Since the values W_i are independently and uniformly distributed over $\{0, 1\}^n$, the success probability of the adversary is at least $1 - e^{-1}$.

C Auxiliary Analysis of SoP-based Constructions

C.1 Matching CDY_k Attack on $F_{B_1}^{\text{SoP}}$

Now, we present a matching attack on CDY_k, which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_1}^{\text{SoP}}$. Let (N, M, T) be the challenge tuple. The adversary proceeds as follows.

1. Choose $2^{n/3}$ distinct keys K^i and compute $U_i = E_{K^i}(N)$ for each $i \in [1, 2^{n/3}]$.
2. Choose $2^{n/3}$ distinct hash keys K_h^i and compute $X_i = H(K_h^i, M)$.
3. Search for indices $a, b, c \in [1, 2^{n/3}]$ such that

$$U_a \oplus U_b \oplus X_c = T.$$

4. Output the key tuple (K^a, K^b, K_h^c) .

Since the values U_i and X_j are independently and uniformly distributed over $\{0, 1\}^n$, the success probability of the adversary is at least $1 - e^{-1}$.

C.2 Analysis of the CMT_k Attack on $F_{B_2}^{\text{SoP}}$

For the first output tuple (K, K', K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_2}^{\text{SoP}}(K, K', K_h, N, M) &= E_K(N) \oplus E_{K'}(N \oplus H(K_h, M)) \\ &= E_K(N) \oplus E_{K'}(N \oplus X). \quad (\text{as } X = H(K_h, M)) \end{aligned}$$

For the second output tuple $(K', K, K_h, N \oplus X, M)$, the tag is computed as:

$$\begin{aligned} F_{B_2}^{\text{SoP}}(K', K, K_h, N \oplus X, M) &= E_{K'}(N \oplus X) \oplus E_K(N \oplus X \oplus H(K_h, M)) \\ &= E_{K'}(N \oplus X) \oplus E_K(N). \quad (\text{as } X = H(K_h, M)) \end{aligned}$$

From the above two equations, it is clear that both tuples produce the same tag. Thus, the algorithm presented in Figure 18 leads to a CMT_k attack on $F_{B_2}^{\text{SoP}}$ with a success probability of 1.

C.3 Proof of Theorem 15

Let $\mathcal{A}$ output a tuple $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that:

1. $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$, and
2. $F_{B_2}^{\text{SoP}}(K_1, K_2, K_h, N, M) = F_{B_2}^{\text{SoP}}(K'_1, K'_2, K'_h, N, M)$.

Let $T := E_{K_1}(N) \oplus E_{K_2}(N \oplus H(K_h, M))$ and $T' := E_{K'_1}(N) \oplus E_{K'_2}(N \oplus H(K'_h, M))$. The adversary succeeds if $T = T'$. We analyze this event through the following cases.

Case 1: $H(K_h, M) = 0^n$ or $H(K'_h, M) = 0^n$. This occurs with probability at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: $K_h = K'_h$. Let $X := H(K_h, M)$. Then $T = T'$ implies

$$E_{K_1}(N) \oplus E_{K_2}(N \oplus X) = E_{K'_1}(N) \oplus E_{K'_2}(N \oplus X).$$

We consider three sub-cases:

- $K_1 = K'_1$ and $K_2 \neq K'_2$: This requires a collision $E_{K_2}(N \oplus X) = E_{K'_2}(N \oplus X)$, which occurs with probability at most $2q_p^2/2^n$.
- $K_2 = K'_2$ and $K_1 \neq K'_1$: This requires a collision $E_{K_1}(N) = E_{K'_1}(N)$, which also occurs with probability at most $2q_p^2/2^n$.
- $K_1 \neq K'_1$ and $K_2 \neq K'_2$: All four ideal-cipher queries are distinct. In this case the adversary must satisfy

$$E_{K_1}(N) \oplus E_{K_2}(N \oplus X) = E_{K'_1}(N) \oplus E_{K'_2}(N \oplus X),$$

which occurs with probability at most $2q_p^4/2^n$.

Case 3: $K_h \neq K'_h$. We divide this case into two sub-cases.

- (3a) **Hash collision:** If $H(K_h, M) = H(K'_h, M)$, this yields a collision in H , which occurs with probability at most $\text{Adv}_H^{\text{CR}}(\mathcal{A}_{cr})$.
- (3b) **Hash outputs differ:** Let $U = E_{K_1}(N)$, $U' = E_{K'_1}(N)$, $X = H(K_h, M)$, $X' = H(K'_h, M)$, $Z = N \oplus X$, $Z' = N \oplus X'$, $V = E_{K_2}(Z)$, and $V' = E_{K'_2}(Z')$. Then $T = T'$ implies $U \oplus V = U' \oplus V'$. We consider two possibilities.
- (3b1) $K_1 = K'_1$: Then $U = U'$, and we require $E_{K_2}(Z) = E_{K'_2}(Z')$. If the last query is forward, this occurs with probability at most $2q_p^2/2^n$.
If the last query is inverse, the adversary must satisfy $E_{K'_2}^{-1}(V') = Z' = Z \oplus X \oplus X'$. If the final query is an inverse ideal-cipher query, this occurs with probability at most $2q_p^2q_h^2/2^n$. If one of the hash queries is the final query among the four relevant queries, the adversary must find a preimage for a target set of cardinality at most $q_p^2q_h$. Hence, in this case, the success probability is at most $\text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2q_h)$.
- (3b2) $K_1 \neq K'_1$: All four queries (K_1, N, U) , (K'_1, N, U') , (K_2, Z, V) , and (K'_2, Z', V') are distinct. If the final query is forward, then $U \oplus V = U' \oplus V'$ occurs with probability at most $2q_p^4/2^n$. If the final query is inverse, similar reasoning gives a bound of $2q_p^2q_h^2/2^n + \text{Adv}_H^{\text{Pre}}(\mathcal{A}'_{pre}; q_p^2q_h)$.

Summing over all cases yields the stated bound.

C.4 Matching attack for RBT_k on $F_{B_2}^{\text{SoP}}$

We now demonstrate the tightness of the RBT_k bound for $F_{B_2}^{\text{SoP}}$ as established in the preceding theorem. The attack proceeds as follows:

- Fix a key K_h , a nonce N and a message M . Compute $X = H(K_h, M)$.
- Select $2^{n/4}$ distinct values for K_1 , and for each $i \in [1, 2^{n/4}]$, compute $U_i = E_{K_1^i}(N)$.
- Select $2^{n/4}$ distinct values for K_2 , and for each $i \in [1, 2^{n/4}]$, compute $V_i = E_{K_2^i}(N \oplus X)$.
- Find indices $a, b, c, d \in [1, 2^{n/4}]$ such that $U_a \oplus V_b = U_c \oplus V_d$.
- Output the tuple $((K_1^a, K_2^b, K_h), (K_1^c, K_2^d, K_h), N, M)$.

Since each U_i and V_i is sampled uniformly at random and independently, the success probability of the attack is at least $1 - e^{-1}$.

C.5 Matching Attack for CDY on $F_{B_2}^{\text{SoP}}$

We show that the tightness of CDY security is proved in the previous theorem. The adversary uses the following strategy:

1. Choose any U and $2^{n/3}$ distinct $K_1^i \leftarrow \{0, 1\}^n$, for $i \in [1, 2^{n/3}]$, and compute $N_i = E_{K_1^i}^{-1}(U)$.
2. Choose $2^{n/3}$ many distinct K_h^i and compute $X_i = H(K_h^i, M)$ for a fixed M .
3. Choose $2^{n/3}$ many distinct K_2^i and compute $Z_i = E_{K_2^i}^{-1}(T \oplus U)$, for $i \in [1, 2^{n/3}]$.
4. Find $a, b, c \in [1, 2^{n/3}]$ such that $Z_b = N_a \oplus X_c$.
5. Output the context tuple $(K_1^a, K_2^b, K_h^c, N_a, M)$.

As N_i , X_i , and Z_i are all independent and uniformly random in $\{0, 1\}^n$, the success probability is at least $1 - e^{-1}$.

C.6 Matching Attack for CDY_k on $F_{B_2}^{\text{SoP}}$

We show that the tightness of CDY_k security is demonstrated by the following attack strategy:

1. Choose any $K_1 \leftarrow \{0, 1\}^n$, and compute $U = E_{K_1}(N)$.
2. Choose $2^{n/2}$ many distinct K_h^i and compute $X_i = H(K_h^i, M)$, for $i \in [1, 2^{n/2}]$.
3. Choose $2^{n/2}$ many distinct K_2^i and compute $Z_i = E_{K_2^i}^{-1}(T \oplus U)$, for $i \in [1, 2^{n/2}]$.
4. Find $a, b \in [1, 2^{n/2}]$ such that $Z_a = N \oplus X_b$.
5. Output the key tuple (K_1, K_2^a, K_h^b) .

As X_i and Z_i are all independent and uniformly random in $\{0, 1\}^n$, the success probability is at least $1 - e^{-1}$.

C.7 Analysis of the CMT_k Attack on $F_{B_3}^{\text{SoP}}$

For the first output tuple (K, K', K_h, N, M) , the tag is computed as:

$$\begin{aligned} F_{B_3}^{\text{SoP}}(K, K', K_h, N, M) &= \mathbf{E}_K(N) \oplus \mathbf{E}_{K'}(N \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M) \\ &= \mathbf{E}_K(N) \oplus \mathbf{E}_{K'}(N \oplus X) \oplus X, \quad \text{where } X = \mathbf{H}(K_h, M). \end{aligned}$$

For the second output tuple $(K', K, K_h, N \oplus X, M)$, the tag is computed as:

$$\begin{aligned} F_{B_3}^{\text{SoP}}(K', K, K_h, N \oplus X, M) &= \mathbf{E}_{K'}(N \oplus X) \oplus \mathbf{E}_K(N \oplus X \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M) \\ &= \mathbf{E}_{K'}(N \oplus X) \oplus \mathbf{E}_K(N) \oplus X. \end{aligned}$$

From the above two expressions, it is clear that both tuples produce the same tag. Thus, the algorithm presented in Figure 20 leads to a CMT_k attack on $F_{B_3}^{\text{SoP}}$ with a success probability of 1.

C.8 Proof of Theorem 18

Let $\mathcal{A}$ output a tuple $((K_1, K_2, K_h), (K'_1, K'_2, K'_h), N, M)$ such that:

1. $(K_1, K_2, K_h) \neq (K'_1, K'_2, K'_h)$, and
2. $F_{B_3}^{\text{SoP}}(K_1, K_2, K_h, N, M) = F_{B_3}^{\text{SoP}}(K'_1, K'_2, K'_h, N, M)$.

Let us denote:

$$\begin{aligned} \mathbf{T} &:= \mathbf{E}_{K_1}(N) \oplus \mathbf{E}_{K_2}(N \oplus \mathbf{H}(K_h, M)) \oplus \mathbf{H}(K_h, M), \\ \mathbf{T}' &:= \mathbf{E}_{K'_1}(N) \oplus \mathbf{E}_{K'_2}(N \oplus \mathbf{H}(K'_h, M)) \oplus \mathbf{H}(K'_h, M). \end{aligned}$$

The adversary succeeds if $\mathbf{T} = \mathbf{T}'$. We analyze the success probability across the following cases.

Case 1: $\mathbf{H}(K_h, M) = 0^n$ or $\mathbf{H}(K'_h, M) = 0^n$. The probability of this event is at most $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}_{\text{pre}}; 1)$.

Case 2: $K_h = K'_h$. We can write:

$$F_{B_3}^{\text{SoP}}(K_1, K_2, K_h, N, M) = F_{B_2}^{\text{SoP}}(K_1, K_2, K_h, N, M) \oplus \mathbf{H}(K_h, M),$$

and similarly for the primed keys. So $\mathbf{T} = \mathbf{T}'$ implies:

$$F_{B_2}^{\text{SoP}}(K_1, K_2, K_h, N, M) = F_{B_2}^{\text{SoP}}(K'_1, K'_2, K_h, N, M),$$

This reduces to key robustness for $F_{B_2}^{\text{SoP}}$. From the RBT_k proof of $F_{B_2}^{\text{SoP}}$, the adversary's success probability in this case is at most $(4q_p^2 + 2q_p^4)/2^n$.

Case 3: $K_h \neq K'_h$. We split this into two sub-cases.

- (a) **Hash collision:** If $\mathbf{H}(K_h, M) = \mathbf{H}(K'_h, M)$, then this is a collision in the random oracle. The probability of this is at most $\mathbf{Adv}_H^{\text{CR}}(\mathbf{Adv}_H^{\text{CR}})$.
- (b) **Hash outputs differ:** Let us define, $U = \mathbf{E}_{K_1}(N)$, $U' = \mathbf{E}_{K'_1}(N)$, $X = \mathbf{H}(K_h, M)$, $X' = \mathbf{H}(K'_h, M)$, $Z = N \oplus X$, $Z' = N \oplus X'$, $V = \mathbf{E}_{K_2}(Z)$ and $V' = \mathbf{E}_{K'_2}(Z')$. The condition $\mathbf{T} = \mathbf{T}'$ implies:

$$U \oplus V \oplus X = U' \oplus V' \oplus X'.$$

We now consider further sub-cases:

- (i) Suppose the hash query for $X' = \mathbf{H}(K'_h, M)$ is the last among all primitive queries. The adversary succeeds if the relation

$$X' = N \oplus Z' \quad \wedge \quad X' = U \oplus V \oplus X \oplus U' \oplus V'$$

is satisfied, and the corresponding components $U = \mathbf{E}_{K_1}(N)$, $V = \mathbf{E}_{K_2}(Z)$, $U' = \mathbf{E}_{K'_1}(N')$, and $V' = \mathbf{E}_{K'_2}(Z')$ have been previously queried.

The probability of this happening is at most $\mathbf{Adv}_H^{\text{Pre}}(\mathcal{A}'_{\text{pre}}; q_p^2)$.

- (ii) Suppose $K_1 = K'_1$, which implies $U = U'$. Then the adversary succeeds if

$$\begin{aligned} &(\mathbf{E}_{K_2}(Z) \oplus X = \mathbf{E}_{K'_2}(Z') \oplus X') \wedge (Z = N \oplus X) \wedge (Z' = N \oplus X') \\ \implies &(\mathbf{E}_{K_2}(Z) \oplus \mathbf{E}_{K'_2}(Z') = X \oplus X') \wedge (Z \oplus Z' = X \oplus X') \\ \implies &\mathbf{E}_{K_2}(Z) \oplus Z = \mathbf{E}_{K'_2}(Z') \oplus Z'. \end{aligned}$$

The success probability of the adversary in this case is at most $2q_p^2/2^n$.

(iii) Suppose $K_1 \neq K'_1$. Then, the adversary succeeds if

$$U \oplus V \oplus X = U' \oplus V' \oplus X'.$$

This implies all four primitive queries are distinct, and also the hash calls yielding X and X' are distinct. We analyze the different possibilities depending on which of the involved ideal cipher queries is the last.

If the last query is for $U' = E_{K'_1}(N)$:

- If this is an inverse query, then for the adversary to succeed, the response must collide with a previously obtained input value. Since at most q_p ideal cipher queries have been made, the success probability is at most $2q_p^2/2^n$.
- If this is a forward query, then U' must satisfy:

$$E_{K'_1}(N) = U \oplus V \oplus X \oplus V' \oplus X'.$$

The values U , (V, X) , and (V', X') can each come from at most q_p previous queries, so the adversary's success probability is at most $2q_p^4/2^n$.

If the last query is for $V' = E_{K'_2}(Z')$:

- If this is an inverse query, then V' must satisfy:

$$E_{K'_2}^{-1}(V') = Z' = N \oplus X'.$$

Since N must be a value from a previous ideal cipher query and X' must match a previously returned hash output, the number of combinations the adversary can match is at most $q_p \cdot q_h$. Hence, the success probability is at most $2q_p^2q_h/2^n$.

- If this is a forward query, then V' must satisfy:

$$E_{K'_2}(Z') = U \oplus V \oplus X \oplus U' \oplus X'.$$

Again, each of U , (V, X) , and (U', X') can be drawn from q_p queries. Thus, the success probability is bounded by $2q_p^4/2^n$.

Combining all the above cases, the result holds.

C.9 Matching Attack for CDY_k on $F_{B_3}^{\text{SoP}}$

Now, we present a matching attack on CDY_k , which establishes that the bound is tight for both CDY_k and CDY security of $F_{B_3}^{\text{SoP}}$. Consider any challenge tuple (N, M, T) , then the adversary proceeds as follows:

1. Choose $2^{n/2}$ many distinct $K_1^i \leftarrow \{0, 1\}^n$, and compute $U_i = E_{K_1^i}(N)$.
2. Choose any K_h and compute $X = H(K_h, M)$.
3. Choose $2^{n/2}$ many distinct $K_2^i \leftarrow \{0, 1\}^n$ and compute $V_i = E_{K_2^i}(N \oplus X)$ for $i \in [1, 2^{n/2}]$.
4. Find $a, b \in [1, 2^{n/2}]$ such that $U_a \oplus V_b = T \oplus X$.
5. Output the key tuple (K_1^a, K_2^b, K_h) .

Since U_i and V_i are independent and uniformly distributed over $\{0, 1\}^n$, and T is also uniformly random, the probability of finding a match is at least $1 - e^{-1}$ by the birthday bound. Therefore, the adversary succeeds with high probability, violating CDY_k security.

D Auxiliary Analysis of nEHtM construction

D.1 RBT_k Attack

We describe a key-robustness attack against the nEHtM construction. The idea is to find two distinct key tuples that produce the same tag for a fixed nonce-message pair. The complete attack strategy is described in Figure 33.

```

1: Select any  $K_h \in \{0, 1\}^{\kappa_h}, M \in \{0, 1\}^*, N \in \{0, 1\}^{n-1}$ ;
2: compute  $X = H_{K_h}(M), Z = N \oplus X$ ;
3: Select  $2^{n/2}$  distinct keys  $K^i \in \{0, 1\}^\kappa, \forall i \in [2^{n/2}]$ ;
4: Compute  $U_i = E(K^i, 0 \| N), \forall i \in [2^{n/2}]$  and  $V_i = E(K^i, 1 \| Z), \forall i \in [2^{n/2}]$ ;
5: Find  $i \neq j \in [2^{n/2}]$  s.t.  $U_i \oplus V_i = U_j \oplus V_j$ ;
6: Output the tuples  $(K^i, K_h, N, M)$  and  $(K^j, K_h, N, M)$ ;

```

Fig. 33: RBT_k adversary against nEHtM

Success Probability. Since the U_i 's are sampled independently and uniformly at random, and the V_i 's are also sampled independently and uniformly at random, and $0 \| N \neq 1 \| Z$, we can apply the birthday paradox over $2^{n/2}$ values. Thus, the probability of finding a collision of the form:

$$U_i \oplus V_i = U_j \oplus V_j, \quad \text{with } i \neq j,$$

is at least $1 - e^{-1}$.

Correctness of the Attack. For both key tuples, the tag produced by nEHtM is the same:

$$\begin{aligned}
\text{nEHtM}(K^i K_h, N, M) &= E(K^i, 0 \| N) \oplus E(K^i, 1 \| Z) \\
&= U_i \oplus V_i, \\
\text{nEHtM}(K^j, K_h, N, M) &= E(K^j, 0 \| N) \oplus E(K^j, 1 \| (N \oplus H_{K_h}(M))) \\
&= U_j \oplus V_j.
\end{aligned}$$

By construction, these are equal. Hence, the adversary successfully finds two different key tuples that result in the same tag, breaking key robustness with probability at least $1 - e^{-1}$.

D.2 Matching Attack for CDY_k

We describe an CDY_k attack against the nEHtM construction for any context selector S . The idea is to find key pair (K, K_h) for S 's output (N, M, T) such that $\text{nEHtM}(K, K_h, N, M) = T$. The complete attack strategy is described in Figure 34.

```

1: Select  $2^{n/2}$  distinct keys  $K_h^i \in \{0, 1\}^{\kappa_h}$ , for all  $i \in [2^{n/2}]$ ;
2: Compute  $X_i = H(K_h^i, M)$ , for all  $i \in [2^{n/2}]$ ;
3: Select  $2^{n/2}$  distinct keys  $K^i \in \{0, 1\}^\kappa$ , for all  $i \in [2^{n/2}]$ ;
4: Compute  $U_i = E_{K^i}(0 \| N)$  for all  $i \in [2^{n/2}]$ ;
5: Compute  $b_i \| Z_i = E^{-1}(K^i, U_i \oplus T)$ , for all  $i \in [2^{n/2}]$ ;
6: Find  $i, j \in [2^{n/2}]$  such that:  $(N \oplus Z_i = X_j) \wedge (b_i = 1)$ ;
7: Output  $(K^i, K_h^j)$ ;

```

Fig. 34: CDY_k adversary against nEHtM

Success Probability. Under the ideal cipher assumption for E the values Z_i , and X_j are sampled independently and uniformly at random from $\{0, 1\}^{n-1}$. Additionally, the bits b_i are independently and uniformly distributed in $\{0, 1\}$, and are independent of Z_i .

Therefore, for each pair $(i, j) \in [2^{n/2}]^2$, the probability that $(N \oplus Z_i = X_j) \wedge (b_i = 1)$ holds is $1/(2 \cdot 2^{n-1}) = 1/2^n$. Hence, the probability that at least one such pair exists is at least $1 - e^{-1}$.

Correctness of the Attack. If the adversary finds such indices i, j satisfying the condition above, then the computed tag is:

$$\begin{aligned} \text{nEHtM}(K^i, K_h^j, N, M) &= \text{E}(K^i, 0 \| N) \oplus \text{E}(K^i, 1 \| (N \oplus \text{H}(K_h^j, M))) \\ &= U \oplus \text{E}(K_2^i, 1 \| (N \oplus X_j)) \\ &= U \oplus \text{E}(K_2^i, 1 \| Z_i) = U \oplus U \oplus T = T. \end{aligned}$$

Therefore, the adversary successfully outputs a valid key tuple for any context selector S with probability at least $1 - e^{-1}$.

E Auxiliary results related to PHash based MACs

E.1 CMT_k Attack on PMAC_Plus

Now, we present a CMT_k attack on PMAC_Plus. The complete attack strategy is described in Figure 35.

- 1 : Choose arbitrary $K_1, K_2, K_3 \in \{0, 1\}^\kappa$ such that $K_1 \neq K_2 \neq K_3$;
- 2 : Choose arbitrary $M_1, M_2 \in \{0, 1\}^n$ and set $M = M_1 \| M_2$;
- 3 : Generate $T = \text{PMAC_Plus}(K_1, K_2, K_3, M)$;
- 4 : Select arbitrary $U \in \{0, 1\}^n$;
- 5 : Compute $V = T \oplus U$;
- 6 : Choose arbitrary $K'_1, K'_2, K'_3 \in \{0, 1\}^\kappa$ such that $K'_1 \neq K'_2 \neq K'_3$;
- 7 : Compute $P = \text{E}_{K'_2}^{-1}(U)$;
- 8 : Compute $Q = \text{E}_{K'_3}^{-1}(V)$;
- 9 : Solve for the system of equations: $X_1 \oplus X_2 = P$ & $X_1 \oplus (2 \otimes X_2) = Q$;
- 10 : Compute $\Delta_0 = \text{E}_{K'_1}(0^n)$, and $\Delta_1 = \text{E}_{K'_1}(1^n)$;
- 11 : Compute $\alpha_1 = ((2 \otimes \Delta_0) \oplus (2^2 \otimes \Delta_1))$ and $\alpha_2 = ((2^2 \otimes \Delta_0) \oplus (2^4 \otimes \Delta_1))$;
- 12 : Compute $M'_1 = \text{E}_{K'_1}^{-1}(X_1) \oplus \alpha_1$ and $M'_2 = \text{E}_{K'_1}^{-1}(X_2) \oplus \alpha_2$;
- 13 : Set $M' = M'_1 \| M'_2$;
- 14 : Output the tuples (K_1, K_2, K_3, M) and (K'_1, K'_2, K'_3, M') ;

Fig. 35: CMT_k adversary against PMAC_Plus

Analysis of the attack. Here, the main step is to solve the system of equations:

$$\begin{aligned} X_1 \oplus X_2 &= P, \\ X_1 \oplus (2 \otimes X_2) &= Q. \end{aligned}$$

This system of equations has a unique solution as the corresponding coefficient matrix has rank 2. The tag computation for the tuple (K'_1, K'_2, K'_3, M') is:

$$\begin{aligned} \text{PMAC_Plus}(K'_1, K'_2, K'_3, M') &= \text{E}_{K'_2}(\text{E}_{K'_1}(\alpha_1 \oplus M'_1) \oplus \text{E}_{K'_1}(\alpha_2 \oplus M'_2)) \\ &\quad \oplus \text{E}_{K'_3}(\text{E}_{K'_1}(\alpha_1 \oplus M'_1) \oplus (2 \otimes \text{E}_{K'_1}(\alpha_2 \oplus M'_2))) \\ &= \text{E}_{K'_2}(X_1 \oplus X_2) \oplus \text{E}_{K'_3}(X_1 \oplus (2 \otimes X_2)) \\ &= \text{E}_{K'_2}(P) \oplus \text{E}_{K'_3}(Q) = U \oplus V = T. \end{aligned}$$

Thus, we have shown a valid CMT_k attack against the PMAC_Plus construction with success probability 1.

E.2 CMT_k and CDY Analysis of 4F-PHash-DbHtF

E.2.1 CMT_k attack on 4F-PHash-DbHtF Let $\pi_1, \pi_2, \pi_3, \pi_4$ be any four independent random permutations. We present a CMT_k attack on 4F-PHash-DbHtF $[\pi_1, \pi_2, \pi_3, \pi_4]$. The complete attack strategy is described in Figure 36.

```

1 : Choose arbitrary  $K \in \{0, 1\}^\kappa$ ;
2 : Choose arbitrary  $M \in \{0, 1\}^n$ ;
3 : Generate  $T = \text{4F-PHash-DbHtF}[\pi_1, \pi_2, \pi_3, \pi_4](K, M)$ ;
4 : Select arbitrary  $R \in \{0, 1\}^n$ ;
5 : Compute  $P \parallel Q = \text{4F}^{-1}(T \parallel R)$ ;
6 : Choose arbitrary  $K' \in \{0, 1\}^\kappa$ ;
7 : Solve for the system of equations:  $X_1 \oplus X_2 = P$  &  $X_1 \oplus (2 \otimes X_2) = Q$ ;
8 : Compute  $\Delta_0 = E_{K'}(0^n)$ , and  $\Delta_1 = E_{K'}(1^n)$ ;
9 : Compute  $\alpha_1 = ((2 \otimes \Delta_0) \oplus (2^2 \otimes \Delta_1))$  and  $\alpha_2 = ((2^2 \otimes \Delta_0) \oplus (2^4 \otimes \Delta_1))$ ;
10 : Compute  $M'_1 = E_{K'}^{-1}(X_1) \oplus \alpha_1$  and  $M'_2 = E_{K'}^{-1}(X_2) \oplus \alpha_2$ ;
11 : Set  $M' = M'_1 \parallel M'_2$ ;
12 : Output the tuples  $(K, M)$  and  $(K', M')$ ;

```

Fig. 36: CMT_k adversary against 4F-PHash-DbHtF

Analysis of the attack. Here, the main step is to solve the system of equations:

$$\begin{aligned} X_1 \oplus X_2 &= P, \\ X_1 \oplus (2 \otimes X_2) &= Q. \end{aligned}$$

This system of equations has a unique solution as the corresponding coefficient matrix has rank 2. Thus, we have shown a valid CMT_k attack against the 4F-PHash-DbHtF construction with success probability 1

E.2.2 CDY Attack on 4F-PHash-DbHtF For the CDY attack against the 4F-PHash-DbHtF construction under any context selector S , the goal is to identify a context tuple (K, M) that produces the target tag T (i.e., the output of S). One can easily find a context tuple (K, M) for a given tag T following the steps 4–11 of Figure 36.